

Systematic screening of brain neurotransmitter enzymes for radical pair mechanism competence reveals quantitative constraints on magnetic field sensitivity

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The radical pair mechanism (RPM) is experimentally validated in avian magnetoreception via cryptochrome, but whether RPM-competent sites exist more broadly among brain enzymes remains unknown. Here we screen nine phosphorus-containing human brain proteins computationally, assigning cofactor-specific ³¹P hyperfine coupling constants, and find that eight satisfy all seven necessary *spin-parameter* conditions for RPM operation under physiological conditions (50 μ T, 310 K), though these conditions are necessary but not sufficient—the exchange coupling J and radical pair generation mechanism impose additional, often prohibitive, constraints. The dominant discriminant is the ³¹P nuclear spin coherence time, which scales as the sixth power of the electron–phosphorus distance, creating a 64-fold gap between FAD enzymes (160 μ s) and PLP enzymes (2.5 μ s). We identify a complementary trade-off: enzymes with strong hyperfine coupling exhibit fast singlet–triplet mixing but short electron coherence, and vice versa. Density matrix simulation of an 8-spin model predicts a magnetic field effect of -4.4% on the singlet yield of MAO-A-catalysed serotonin oxidation at $J = 0$ (an upper bound; the 8-spin truncation itself overestimates MFE by $\sim 20\text{--}40\%$); the actual magnitude is negligible at the crystallographic radical pair distance ($|J| \sim 500\text{--}1000$ MHz). These findings establish that spin-parameter prerequisites for RPM are broadly distributed across brain enzymes, but simultaneously reveal that the exchange coupling constraint ($|J| < 10$ MHz required), homeostatic attenuation (7-fold; Table 6), and CASSCF evidence against SET in MAO-A (Fig. 7) collectively impose severe quantitative barriers. Cryptochrome—the only enzyme with a confirmed, spatially separated radical pair—emerges as the sole currently defensible RPM host, with a predicted circadian period shift of 2.7–8.2 minutes at Earth’s field. The primary contribution is a falsifiable framework with experimentally testable predictions, not a claim that RPM operates in neurotransmitter-metabolising enzymes.

Quantum biology has advanced rapidly over the past two decades, with proposed quantum effects in photosynthetic energy transfer[1, 2] (though the functional role of observed coherences remains debated), established hydrogen tunnelling in enzymes[3], and avian magnetoreception[4–6]. Among these, the radical pair mechanism (RPM) stands out as the most experimentally validated: transient radical pairs generated by electron transfer undergo coherent singlet–triplet interconversion driven by hyperfine coupling (HFC) to magnetic nuclei, producing reaction yields that are sensitive to weak magnetic fields[7, 8]. The RPM in cryptochrome has been confirmed to respond to Earth-strength fields[6, 9], and recent work has invoked the quantum Zeno effect to explain the sensitivity of tightly bound radical pairs[10]. In parallel, Fisher proposed that ³¹P nuclear spins in Posner molecules could serve as neural qubits with long coherence times[11], a hypothesis that has stimulated considerable theoretical work[12] but remains experimentally unverified.

Previous proposals for quantum effects in the brain—including Penrose and Hameroff’s orchestrated objective reduction in microtubules and Fisher’s Posner molecule hypothesis[11]—have thus far lacked reproducible experimental confirmation, and the field rightly demands that extraordinary claims be supported by quantitative, testable predictions grounded in established physics. The RPM is distinct from these proposals in that it is experimentally validated in non-biological[8] and biological[6] systems, and its physical requirements are well understood. Despite this strong foundation, and while Zadeh-Haghighi & Simon[40] have reviewed the broad applicability of RPM across biology and individual studies have examined MFEs in specific flavoenzymes[5, 22] and mitochondrial systems[23], a systematic *comparative* assessment of RPM competence across brain enzymes using cofactor-specific spin parameters

and explicit brain-condition corrections has not been performed. The brain contains dozens of enzymes with phosphorus-containing cofactors—flavin adenine dinucleotide (FAD), pyridoxal 5'-phosphate (PLP), and nicotinamide adenine dinucleotide phosphate (NADPH)—each containing ^{31}P nuclei (spin-1/2, 100% natural abundance) whose HFCs to the radical centre range from ~ 2 MHz (remote phosphates) to ~ 200 MHz (the FMN-phosphate in FAD[13]), depending on the through-bond pathway. Monoamine oxidases catalyse the oxidative deamination of serotonin and dopamine. The catalytic mechanism has been debated: stopped-flow kinetics[24] and substrate analogue studies support a single-electron transfer (SET) pathway that would generate transient radical pair intermediates, while computational studies have also proposed concerted (polar) mechanisms in which no radical pair forms. An alternative radical partner to the substrate amine could be superoxide ($\text{O}_2^{\bullet-}$), which is generated during MAO catalysis and has been proposed as the magnetosensitive radical partner in cryptochrome[21]; a flavin-superoxide pair would have different spin dynamics from the flavin-amine pair modelled here, potentially with larger RP separation and weaker exchange coupling, which would be more favourable for RPM. We note that direct spectroscopic evidence for radical pair intermediates in MAO catalysis—such as spin-trapping or transient EPR—has not been reported, and this constitutes a key uncertainty in our analysis. Nonetheless, MAO-A is a particularly intriguing RPM candidate given its central role in serotonin metabolism. Without a quantitative framework for comparing RPM competence across brain proteins, the question of whether quantum spin effects influence neurotransmitter metabolism cannot be addressed.

We screened nine representative brain proteins (Table 1), selected to span all three major phosphorus-cofactor classes (4 FAD, 2 PLP, 1 NADPH, 1 NADH, 1 PLP+metal) and include both a validated RPM host (CRY, positive control) and a transition-metal enzyme (SRR, expected negative control). While the brain contains >50 phosphorus-cofactor enzymes, the dominant physical discriminant we identify—the e^- -P distance, which takes only three discrete values across cofactor classes (3.5 Å for PLP, 7 Å for FAD-FMN-P, 12 Å for AMP-P)—means that the tier structure is determined by cofactor type rather than by individual protein identity. Additional enzymes within each class would populate the existing tiers without creating new ones (Fig. 2b); we therefore prioritised cofactor-class coverage over exhaustive enumeration. The pipeline and all code are released for extension to the full enzyme set. Using GFN2-xTB[16], we combined three contributions: (i) a multi-condition RPM assessment under brain-relevant conditions; (ii) the identification of nuclear $T_2(^{31}\text{P})$ —governed by the r^{-6} PRE distance scaling[18] and thus robust to electronic-structure uncertainties—as the dominant discriminant; and (iii) density matrix spin dynamics yielding testable MFE predictions. For each quantum-active site, we computed electronic TDMs, oscillator strengths, HFCs, SOC, and singlet-triplet gaps. All nine proteins exhibit TDMs of 6.4–11.1 D and oscillator strengths $f_{\text{osc}} = 0.04$ –0.58. FAD enzymes show the largest TDMs (~ 10 D) owing to the extended isoalloxazine π -system, while PLP enzymes have moderate TDMs (6–9 D). SOC values range from 63 cm^{-1} (GAD67, MAO-A) to 382 cm^{-1} (SRR, due to Mn^{2+}).

We assessed each protein against seven necessary RPM conditions. These conditions function as binary filters rather than differentiators: their purpose is to exclude physically disqualified candidates (such as transition-metal enzymes), while the Q_{RPM} metric (Eq. 3) provides the quantitative ranking among those that pass. The conditions are: radical pair generation capacity (C1: for photoenzymes such as CRY, $f_{\text{osc}} > 0.01$ ensures efficient light-driven RP formation; for dark enzymes such as MAO-A, we instead require a thermodynamically accessible single-electron transfer step, assessed by the vertical ionisation potential difference $|\text{IP}_{\text{sub}} - \text{EA}_{\text{cof}}| < 2\text{ eV}$ —a deliberately permissive threshold chosen to avoid false negatives at the screening stage, noting that the SET mechanism itself remains debated[24]); ^{31}P nuclear spins (C2, $n_{\text{P}} \geq 1$); electron spin coherence (C3, $T_2^e > 0.1\text{ ns}$, ensuring at least one full HFC-driven S-T oscillation); HFC exceeding electron Zeeman at Earth's field (C4, $A/\omega_S > 10$, the strong-coupling condition); $T_1^e \gg T_2^e$ (C5, maintaining spin polarisation during S-T evolution); nuclear coherence exceeding radical pair lifetime (C6, $T_2(^{31}\text{P}) > 1\text{ }\mu\text{s}$); and moderate SOC (C7, $10 < \xi < 200\text{ cm}^{-1}$, permitting ISC without destroying coherence). That 8/9 proteins pass reflects the physical reality that organic-cofactor enzymes generically satisfy RPM prerequisites; the non-trivial result is the >100 -fold quantitative variation in Q_{RPM} among qualifying candidates. As shown in Fig. 1, eight proteins satisfy 7/7 conditions. Serine racemase (SRR) fails three conditions owing to its Mn^{2+} centre (SOC = 382 cm^{-1} ,

$T_2^e \approx 0$, no active-site ^{31}P), consistent with the expectation that transition-metal-containing enzymes are strongly disfavoured for RPM operation owing to rapid spin relaxation[8].

To discriminate among qualifying candidates, we decomposed T_2^e into five contributions: inhomogeneous HFC dephasing (T_2^* , Anderson–Weiss[17]), dipolar coupling (T_2^{dd}), g -anisotropy modulation ($T_2^g \propto \Delta g^2 \omega_0^2 \tau_c$), HFC secular modulation (T_2^A), and SOC-induced relaxation. A critical finding emerges at Earth’s field: because $T_2^g \propto \omega_0^2$ and ω_0 drops by 6.7×10^3 from X-band to 50 μT , the g -anisotropy channel effectively vanishes (Fig. 2a). With cofactor-specific HFCs, the remaining HFC modulation channel ($1/T_2^A \propto \sum A_i^2$) now differentiates strongly: FAD enzymes with two ^{31}P at 200 MHz retain $T_2^e \approx 1.1$ ns, while PLP enzymes (one ^{31}P at ~ 100 MHz) reach $T_2^e \approx 3\text{--}5$ ns at Earth’s field—the *longest* electron coherence among all candidates. This reversal—PLP enzymes have the shortest nuclear T_2 but the longest electron T_2 —is a direct consequence of the $\sum A_i^2$ dependence of the HFC modulation rate[17], applied here to the specific molecular geometries of brain enzyme cofactors.

The differentiating variable instead proves to be nuclear spin coherence. The ^{31}P T_2 in the coupled electron–nuclear system is dominated by paramagnetic relaxation enhancement (PRE)[18], scaling as r^{-6} where r is the electron–phosphorus distance. For PLP enzymes ($r \approx 3.5$ Å, phosphorus bonded directly to the chromophore), we compute $T_2(^{31}\text{P}) = 2.5$ μs ; for FAD enzymes ($r \approx 7$ Å), $T_2(^{31}\text{P}) = 160$ μs —a 64-fold difference arising from $(7/3.5)^6 = 64$ (Fig. 2b). In the diamagnetic state, ^{31}P relaxation enters the extreme-narrowing regime at Earth’s field ($\omega_0 \tau_c \ll 1$), yielding $T_1 = T_2 = 3.2$ ms regardless of cofactor—a timescale that would, in principle, permit nuclear spins to retain non-equilibrium polarisation between successive enzymatic turnovers ($k_{\text{cat}}^{-1} \sim 100$ ms), although this requires that the polarisation survives the conformational changes and substrate rebinding events of each catalytic cycle, which has not been demonstrated.

At 50 μT , the ^{31}P HFC ($A = 200$ MHz) vastly exceeds the electron Larmor frequency ($\omega_S = 1.4$ MHz), placing the system in the strong-coupling limit (Fig. 3a). The mixing angle between $|\alpha_e \beta_n\rangle$ and $|\beta_e \alpha_n\rangle$ approaches $\pi/4$, causing the nominally forbidden zero-quantum (ZQ) transition to acquire a magnetic TDM of $1.22 \mu_B$ —70% of the allowed EPR value ($1.73 \mu_B$). This near-complete mixing is the physical basis for efficient ^{31}P -driven singlet–triplet interconversion at geomagnetic field strengths, a regime qualitatively different from laboratory EPR conditions.

To quantify the MFE for the top-ranked candidate as a case study, we simulated an eight-spin radical pair model of the MAO-A active site ($2e + 2 \times ^{31}\text{P} + 4 \times ^1\text{H}$; Hilbert space dimension 256) by exact diagonalisation of:

$$\hat{H} = J \mathbf{S}_1 \cdot \mathbf{S}_2 + \sum_{i \in \text{rad } 1} A_i \mathbf{S}_1 \cdot \mathbf{I}_i + \sum_{j \in \text{rad } 2} A_j \mathbf{S}_2 \cdot \mathbf{I}_j + g\mu_B B (\hat{S}_{1z} + \hat{S}_{2z}) \quad (1)$$

with cofactor-specific HFCs ($A(^{31}\text{P}_{\text{FMN}}) = 200$ MHz[13], $A(^{31}\text{P}_{\text{AMP}}) = 2$ MHz, $A(^1\text{H}) = 2.7$ MHz; see Table 3), where the two ^{31}P and two ^1H nuclei are assigned to radical 1 (flavin) and the remaining two ^1H to radical 2 (amine), and recombination rate $k = 1$ MHz (corresponding to a RP lifetime of ~ 1 μs , consistent with flavin RP lifetimes of 0.1–10 μs measured by transient absorption in related flavoproteins[5]). The exchange coupling was set to $J = 0$. This is the central assumption of the simulation and warrants careful scrutiny. For CRY, radical pair separations of 10–20 Å along the Trp chain yield $|J| < 1$ MHz, and $J = 0$ is well justified[15]. For MAO-A, the FAD N5–substrate amine distance is ~ 3.5 Å in crystal structures[14], and the exchange coupling at this distance is estimated at $|J| \sim 100\text{--}1000$ MHz based on the exponential distance dependence $J = J_0 \exp(-\beta r)$ with $\beta \approx 2.8 \text{ Å}^{-1}$ established for through-space exchange in organic radical pairs[8]. Our systematic scan over $|J| = 0\text{--}1000$ MHz (Fig. 6) reveals non-monotonic behaviour: MFE is -4.4% at $J = 0$, reaches a level anti-crossing resonance near $|J| \approx 5$ MHz (where the $S\text{--}T_+$ or $S\text{--}T_-$ crossing amplifies spin mixing), then declines to -3% at $|J| = 100$ MHz and -0.006% at $|J| = 1000$ MHz. For the MAO-A active site ($r \approx 3.5$ Å, $|J| \sim 500\text{--}1000$ MHz), MFE is negligible ($< 0.1\%$); the electron–electron dipolar coupling ($D \approx 5000$ MHz at 3.5 Å) would further suppress S-T mixing, reinforcing this conclusion. We therefore conclude that RPM-based magnetosensitivity in MAO-A is only viable if the radical pair achieves spatial separation $r > 8$ Å ($|J| < 10$ MHz) during the catalytic cycle—for example, through substrate dissociation or

a conformational gate. No transient EPR or spin-trapping evidence currently exists for a long-lived, separated radical pair intermediate in MAO catalysis[14], and obtaining such evidence is a prerequisite for the hypothesis. In CRY, by contrast, the Trp-triad/tetrad pathway produces RPs with $r = 15\text{--}20 \text{ \AA}$ and $|J| < 1 \text{ MHz}$, placing it firmly in the MFE-competent regime[15]. Assuming the RP is born in a pure singlet state ($\rho_0 = |S\rangle\langle S| \otimes \rho_{\text{nuc}}$ with uniform nuclear spin density) and that J in Eq. 1 represents the isotropic Heisenberg exchange coupling (the anisotropic dipolar contribution is $< 1 \text{ MHz}$ at $r > 3.5 \text{ \AA}$ and is neglected), we computed the singlet yield analytically:

$$\Phi_S(B) = k \sum_{m,n} \frac{[\tilde{P}_S]_{mn} [\tilde{\rho}_0]_{nm}}{k + i(E_m - E_n)} \quad (2)$$

where tildes denote the energy eigenbasis and k enters as a uniform singlet/triplet recombination rate in the Haberkorn formalism[28] (derivation in Supplementary Section 1, Eqs. S14–S16). We note that alternative recombination models (e.g., the Jones–Hore approach) can modify the absolute MFE by up to $\sim 50\%$ while preserving the field dependence[22]; our values should be interpreted within this methodological uncertainty. The MFE curve (Fig. 4) peaks at -4.4% near $B = 0.9 \text{ mT}$, with -0.3% at Earth’s field ($50 \text{ }\mu\text{T}$). These values are consistent with experimental MFEs of $1\text{--}15\%$ observed in flavoenzyme reactions[5].

We ranked all candidates using a heuristic composite metric evaluated under brain conditions. We stress that Q_{RPM} is not derived from first principles; it is a dimensioned product of five independently estimated quantities chosen because each is a necessary—but not sufficient—condition for RPM. Its value lies in identifying which candidates fail dramatically (SRR) and which pass all conditions simultaneously, not in the precise numerical ranking:

$$Q_{\text{RPM}} = f_{\text{osc}} \times T_2^e(\text{ns}) \times n_{\text{P}}^{\text{eff}} \times \frac{T_2(^{31}\text{P})}{1 \text{ }\mu\text{s}} \times \left(\frac{100}{\xi}\right)^2 \quad (3)$$

where f_{osc} serves as a proxy for RP generation efficiency (for photoenzymes, it quantifies light absorption; for dark enzymes such as MAO-A, it correlates with the π -system extent that stabilises radical intermediates, though this correlation is approximate—see below for J -aware corrections), and $n_{\text{P}}^{\text{eff}}$ is the number of ^{31}P nuclei with $A > 20 \text{ MHz}$ (Table 3), reflecting the finding that only ^{31}P atoms with substantial through-bond coupling to the radical centre contribute to S-T mixing. For most FAD enzymes, $n_{\text{P}}^{\text{eff}} = 1$ (FMN-P only; AMP-P has $A \approx 2 \text{ MHz}$). In the MAO-A crystal structure (PDB 2BXR), the two phosphate groups are nearly equidistant from the radical centre ($\Delta r < 0.2 \text{ \AA}$), causing both to be classified as proximal; we conservatively assign $n_{\text{P}}^{\text{eff}} = 2$ for MAO-A but note that this may overestimate its advantage relative to CRY ($n_{\text{P}}^{\text{eff}} = 1$). For PLP, $n_{\text{P}}^{\text{eff}} = 1$; for NADPH, $n_{\text{P}}^{\text{eff}} = 2$ (NMN-P and 2'-P). An important consequence of the cofactor-specific HFC assignment is that the electron spin coherence time T_2^e increases significantly for proteins whose HFCs were reduced: PLP enzymes gain ~ 3 -fold (from 0.5 to 1.6 ns) and NADPH enzymes gain ~ 3.6 -fold (from 0.23 to 0.85 ns), because HFC secular modulation ($1/T_2^A \propto \sum A_i^2$) is the dominant T_2^e channel.

The ranking (Table 2) reveals a clear tier structure. Tier 1 (FAD enzymes, $Q = 100\text{--}500$) is separated from Tier 2 (PLP and NADPH enzymes, $Q = 10\text{--}100$) by the 64-fold nuclear T_2 gap that depends on crystallographic distances. A notable consequence of the revised HFC values is that PLP enzymes (DDC, GAD67), despite their short nuclear T_2 ($2.5 \text{ }\mu\text{s}$), now exhibit the *longest* electron coherence ($T_2^e \approx 1.6 \text{ ns}$) among all candidates—a direct result of their reduced HFC modulation. Within Tier 1, MAO-A, MAO-B, CRY, and DAO are not statistically distinguishable given the propagated uncertainties (estimated $\pm 50\%$ in Q_{RPM} , arising primarily from $\pm 50\%$ in f_{osc} and $\pm 30\%$ in T_2^e); the tier assignment, not the intra-tier ranking, constitutes our primary conclusion.

Because Q_{RPM} deliberately excludes the exchange coupling J —the single most important determinant of MFE magnitude—the intra-Tier 1 ranking must be interpreted with a critical caveat. When J -dependent MFE is considered (Fig. 6), CRY is the only candidate with established large RP separation ($r = 15\text{--}20 \text{ \AA}$, $|J| < 1 \text{ MHz}$ [15]), placing it unambiguously first for *demonstrated* RPM viability. MAO-A ranks first on Q_{RPM} (spin parameters alone) but ranks below CRY on any J -aware metric because its active-site

RP distance (~ 3.5 Å) yields $|J| \gg 100$ MHz and negligible MFE unless a separated RP intermediate is formed. We therefore present two complementary rankings in Table 2: “ Q_{RPM} ” (spin-parameter ranking, J -agnostic) and “ J -aware” (where CRY is promoted to rank 1 and dark enzymes lacking evidence for separated RPs are flagged).

The identification of MAO-A as the top-ranked candidate has implications for neuroscience. MAO-A is the primary enzyme for serotonin degradation, and its inhibitors are established antidepressants[19]. However, therapeutic MAO inhibition produces $>50\%$ reduction in MAO activity—orders of magnitude larger than any plausible MFE—and we do not claim clinical relevance for the effects predicted here. Our predicted MFE of -4.4% at optimal field is quantitatively consistent with experimental MFEs of 1–15% observed in isolated flavoprotein reactions by Maeda *et al.*[5] and in glucose oxidase ($\sim 30\%$ at 30 mT, a considerably higher field) by Jones *et al.*[22]. Although these experimental MFEs were measured at higher fields than our predicted peak (~ 1 mT), they establish that spin-dependent chemistry in flavoenzymes produces measurable yield changes. We propose three direct experimental tests, noting that our own J -scan analysis predicts the first of these will yield a null result unless a separated RP intermediate exists: (1) measurement of MAO-A catalytic rate (k_{cat} , K_{M}) as a function of static magnetic field (0–50 mT) using a fluorometric serotonin oxidation assay—a null result at <5 mT would constrain the RP separation to $r < 8$ Å and effectively rule out RPM in this enzyme, while a positive result at ~ 1 mT would imply a transiently separated RP; (2) isotope substitution of ^{31}P ($I = 1/2$) with ^{32}P ($I = 0$, spinless) in FAD, which should abolish the ^{31}P -driven component of the MFE—although we note that this experiment requires handling of radioactive ^{32}P (β^- , $t_{1/2} = 14.3$ d) and reconstitution of labelled FAD into apo-MAO-A, which is technically demanding; an alternative is selective deuteration of ^1H near the radical centre, which would modify (though not abolish) the MFE pattern; and (3) measurement of serotonin levels in primary neuronal cultures exposed to static magnetic fields of 0.1–5 mT. We emphasise that the most experimentally accessible prediction is the -4.4% MFE at ~ 1 mT rather than the -0.3% at Earth’s field: the former lies well above typical enzyme assay variability ($\text{CV} \sim 3\text{--}5\%$) and can be tested with standard fluorometric methods[24]. The Earth-field effect, while of greater biological relevance, would require differential measurement techniques (for example, paired measurements with and without magnetic shielding) to resolve.

CRY, the only protein with experimental RPM validation[6], ranks in Tier 1 but not first on Q_{RPM} . However, our CASSCF(2,2)/NEVPT2 analysis (Figs. 7 and 8) reveals that this ranking must be fundamentally re-evaluated. The diradical character y_0 of the lumiflavin–methylamine model is only 0.03–0.09 across the reaction coordinate, and the CASSCF S-T gap (3.2–4.1 eV) is far larger than the xTB estimate. This indicates that the SET mechanism assumed for MAO-A is not supported at this level of theory; polar/concerted pathways are favoured. When RP-generation feasibility is incorporated as a weight ($w = 1.0$ for experimentally confirmed RP, 0.3 for uncertain, 0 for excluded), the effective ranking becomes: (1) CRY ($Q_{\text{eff}} = 391$, confirmed RP), (2) MAO-A ($Q_{\text{eff}} = 137$, uncertain RP), (3) MAO-B ($Q_{\text{eff}} = 115$, uncertain). IDH and LDH, which operate via hydride transfer (no RP), drop to $Q_{\text{eff}} = 0$. This revision places CRY unambiguously first—the physically correct result—while MAO-A remains a conditional candidate pending experimental evidence for a separated RP intermediate.

The preceding MAO-A analysis serves a deliberate purpose: by quantifying RPM for the highest- Q_{RPM} dark enzyme and showing that exchange coupling ($|J| \sim 500\text{--}1000$ MHz), CASSCF-supported polar mechanism ($y_0 < 0.1$), and homeostatic attenuation ($7.3\times$; Table 6) each independently suppress the effect, it establishes that high spin-parameter scores alone are insufficient for functional RPM—RP generation and spatial separation are the decisive bottlenecks.

These findings motivate a three-layer conceptual framework (Fig. 5). Given that our CASSCF analysis identifies CRY as the only enzyme with confirmed RP generation, we ground this framework explicitly in CRY biology, where every link in the chain has experimental support, before discussing extension to other enzymes.

CRY-based three-layer framework. *Layer 1 (nuclear spin quantum memory).* The two ^{31}P nuclei in FAD’s phosphodiester chain carry nuclear spins ($I = 1/2$, 100% abundance) with diamagnetic coherence times of 3.2 ms (extreme-narrowing regime at 50 μT). During CRY’s radical pair lifetime ($\tau_{\text{RP}} \sim 1\text{--}$

10 μ s[5]), the FMN-phosphate ^{31}P ($A = 200$ MHz, $r_{eP} \approx 7$ Å, PRE-limited $T_2 = 160$ μ s) acts as a quantum register: its nuclear spin state modulates the singlet–triplet interconversion rate via HFC, encoding magnetic-field information into the RP’s spin dynamics. After RP recombination and return to the diamagnetic state, the ^{31}P retains its spin orientation for ~ 3.2 ms—long enough to survive between photocycles ($\sim$ ms timescale under physiological illumination).

Layer 2 (radical pair quantum–classical interface). Blue-light absorption ($\lambda_{\text{max}} \approx 450$ nm, $f_{\text{osc}} = 0.27$) generates $[\text{FAD}^{\bullet-} \cdots \text{Trp}^{\bullet+}]$ via sequential electron transfer along the conserved Trp triad/tetrad[6]. The RP separation ($r = 15\text{--}20$ Å) ensures $|J| < 1$ MHz[15], placing the system firmly in the RPM-competent regime. At Earth’s field, ^{31}P -driven S- T_0 mixing (ZQ TDM = $1.22 \mu_B$, 70% of allowed) produces a field-dependent singlet yield $\Phi_S(B)$. An 8-spin simulation of the CRY RP ($2e + ^{31}\text{P}_{\text{FMN}} + 2 \times ^1\text{H}_{\text{FAD}} + 2 \times ^1\text{H}_{\text{Trp-}\beta} + ^{14}\text{N}_{\text{Trp}}$; Fig. 9) predicts MFE = -5.6% at $B = 0.29$ mT and -0.55% at $50 \mu\text{T}$. The larger MFE and lower peak field compared to MAO-A (-4.4% at 0.9 mT) reflect the asymmetric HFC distribution: CRY’s Trp radical carries large ^1H HFCs (20 MHz) absent in the amine radical. At $|J| = 1$ MHz (the upper bound for CRY’s Trp-triad RP[15]), the Earth-field MFE is reduced by only 14% (from -0.94% to -0.80% ; Fig. 9b), confirming that the RPM operates efficiently in CRY’s established J regime. These values are consistent with the $\sim 1\text{--}15\%$ MFEs measured in isolated CRY[5]. The RP’s spin state is thus the transduction element: a quantum degree of freedom (nuclear spin orientation relative to the external field) is converted into a classical observable (product branching ratio).

Layer 3 (classical circadian signalling). CRY functions as a transcriptional repressor in the CLOCK/BMAL1 feedback loop that generates ~ 24 -hr oscillations in gene expression across the suprachiasmatic nucleus (SCN)[38]. An MFE on CRY’s photoreduction yield would modulate the ratio of photoactivated (signalling-competent) to ground-state CRY. Unlike the MAO-A/serotonin pathway, CRY signalling operates via conformational change and protein–protein interaction (CRY $\rightarrow$ PER $\rightarrow$ CLOCK/BMAL1 complex disruption) rather than metabolite concentration; homeostatic buffering of the type that attenuates serotonin perturbations (reuptake, autoreceptors) does not apply. Instead, the circadian system amplifies small input changes through its limit-cycle dynamics: mathematical models of the SCN oscillator show that a sustained $\sim 1\%$ change in CRY activation rate shifts the circadian period by $\sim 5\text{--}15$ minutes[39]. Applying our CRY-specific simulation result (MFE = -0.55% at Earth’s field), the predicted period shift is $\sim 2.7\text{--}8.2$ minutes (Fig. 9c)—within the range of experimentally observed inter-individual variation (~ 30 minutes) and comparable to seasonal photoperiod effects.

CRY three-layer information flow:

$$\underbrace{B_{\text{Earth}} \rightarrow ^{31}\text{P spin}}_{\text{Layer 1}} \rightarrow \underbrace{[\text{FAD}^{\bullet-} \cdots \text{Trp}^{\bullet+}] \rightarrow \Phi_S(B)}_{\text{Layer 2}} \rightarrow \underbrace{\text{CRY}^* \rightarrow \text{PER/CLOCK} \rightarrow \Delta T \approx 3\text{--}8 \text{ min}}_{\text{Layer 3}}$$

This CRY-centred pathway has three decisive advantages over the MAO-A hypothesis: (i) the RP is experimentally confirmed with $|J| < 1$ MHz[6, 15]; (ii) the Layer 2 $\rightarrow$ 3 transduction does not require metabolite concentration changes and is therefore immune to homeostatic buffering; and (iii) CRY’s role in circadian timing provides a plausible functional context—magnetic field sensitivity could entrain or modulate the circadian phase, which is known to influence mood, sleep, and seasonal behaviour. The experimentally observed magnetic sensitivity of *Drosophila* CRY[6] demonstrates that this pathway is functional in at least one organism; whether human CRY retains this capacity is an open question.

Extension to dark enzymes. For MAO-A and other dark enzymes, Layer 2 activation remains the critical bottleneck. If future experiments (transient EPR, CIDEP) demonstrate a separated RP intermediate during MAO catalysis, the framework extends naturally: the information flow becomes $B \rightarrow ^{31}\text{P} \rightarrow [\text{FADH}^{\bullet} \cdots \text{Sub}^{\bullet}] \rightarrow \Phi_S \rightarrow k_{\text{cat}} \rightarrow [5\text{-HT}]$. However, the MAO-A pathway faces three additional constraints absent in CRY: (i) $|J| \sim 500\text{--}1000$ MHz at the active-site distance, requiring RP separation; (ii) homeostatic attenuation reducing $\delta[5\text{-HT}]$ by ~ 7 -fold (Table 6); and (iii) the CASSCF-supported polar mechanism, which would preclude RP formation entirely. We therefore propose that the CRY pathway is the only currently defensible instantiation of the three-layer framework in the human brain.

The potential biological scale of these effects can be estimated, albeit with large uncertainties. Taking an MFE of 0.3% (Earth’s field, $J = 0$ upper bound), $k_{\text{cat}} \approx 10 \text{ s}^{-1}$ [14], and $\sim 10^{10}$ MAO-A

molecules brain-wide, the flux perturbation is $\sim 3 \times 10^8$ molecules s^{-1} , or $\sim 0.3\%$ of the total brain-wide serotonin turnover ($\sim 10^{11}$ molecules s^{-1})—but this estimate spans orders of magnitude (0 to $\sim 10^{10}$) given the unknown J and the order-of-magnitude uncertainty in enzyme copy number. We note that alternative non-RPM mechanisms for biological MFEs—including magnetomechanical torques on anisotropic structures, Lorentz forces on ion currents, and interactions with biogenic magnetite nanoparticles—have been proposed and are not excluded by our analysis; our framework addresses only the RPM channel. Experimental precedent for millitesla-field effects on cellular bioenergetics has been reported[23]. It is worth asking why, if RPM effects exist in neurotransmitter enzymes, they have not been clearly observed despite decades of MFE research on flavoproteins. The most likely answer is that the combination of large exchange coupling ($|J| \gg 100$ MHz at active-site distances), rapid product release, and homeostatic buffering collectively suppress the effect to levels below routine assay sensitivity (CV $\sim 3\text{--}5\%$)—precisely the quantitative constraints our analysis identifies. We note that our framework, like Fisher’s Posner molecule hypothesis[11], involves ^{31}P nuclear spins, but the mechanisms are fundamentally different in both physics and plausibility: Posner clusters invoke entangled free-diffusing phosphate ions (experimentally unverified), whereas our framework concerns enzyme-bound ^{31}P coupled to established radical pair spin dynamics via HFC. No entanglement between separate molecules is required or proposed here.

We stress that our results do not demonstrate quantum effects in neural processing. They establish that (i) the spin-parameter prerequisites for RPM are broadly distributed across brain enzymes, with nuclear T_2 as the dominant discriminant; (ii) RP generation—the critical bottleneck—is confirmed only for CRY and remains uncertain for dark enzymes including MAO-A, where CASSCF analysis favours polar over SET mechanisms; and (iii) even under the most favourable assumptions ($J = 0$), homeostatic attenuation reduces the physiological signal to $\sim 0.6\%$ of the raw MFE. The primary contribution of this work is a quantitative framework that converts the question “could RPM operate in brain enzyme X?” into specific, experimentally testable predictions for each candidate.

Our analysis has several limitations. First, GFN2-xTB provides semi-quantitative accuracy for electronic structure. To assess this, we performed B3LYP/def2-SVP calculations with TD-DFT (TDA) and ΔSCF vertical IP/EA on model compounds—lumiflavin (FAD isoalloxazine, 30 atoms, 312 basis functions) and a PLP-methylamine Schiff base (26 atoms, 269 basis functions)—using PySCF 2.12.1. The comparison (Table 5) reveals systematic errors of similar magnitude but with instructive differences. For the FAD HOMO–LUMO gap, xTB gives 1.42 eV and B3LYP gives 1.26 eV, both substantially below the experimental value of 2.76 eV; however, TD-B3LYP places the bright excitation at 2.29 eV (541 nm, S_4 state), considerably closer to experiment (450 nm) than the Kohn–Sham gap. For f_{osc} , xTB overestimates (0.50 vs. 0.27 experimental, +85%) while TD-B3LYP underestimates (0.089, –67%), bracketing the experimental value. Vertical ionisation potentials show a complementary pattern: xTB overestimates the FAD IP by +2.8 eV (10.78 vs. 8.0 eV experimental) while B3LYP underestimates by –1.4 eV (6.64 eV). For PLP, B3LYP gives a Kohn–Sham gap of 3.04 eV (error –0.83 eV), substantially better than xTB (1.93 eV, error –1.94 eV), and the PLP vertical IP (8.72 eV, B3LYP) agrees with the experimental estimate (~ 8.5 eV) to within 0.2 eV. We verified that substituting DFT-derived f_{osc} values into the Q_{RPM} formula reduces absolute scores by ~ 5.8 -fold but preserves the tier structure and intra-tier ordering (six of eight ranks unchanged; CRY and DAO swap within Tier 1). Crucially, the key differentiator—the 64-fold gap in nuclear $T_2(^{31}\text{P})$ between PLP and FAD enzymes—depends on crystallographic e^- –P distances (3.5 vs. 7 Å), not on computed electronic properties, and is therefore robust to either method’s inaccuracies. Similarly, the spin relaxation times T_2^e , T_1^e , and nuclear T_2 are independent of the gap and f_{osc} (they depend on HFC, SOC, and distances), so the DFT revision leaves all relaxation parameters unchanged. We note that a proper TDDFT benchmark would require a range-separated functional (CAM-B3LYP or $\omega\text{B97X-D}$), a triple- ζ basis, and a continuum solvent model; we present the B3LYP/def2-SVP results as a first-level validation confirming that the tier structure is method-independent.

Second, the introduction of cofactor-specific HFCs reveals that IDH (NADPH) occupies an intermediate position: its nuclear T_2 is long (160 μs , FAD-like distance) but its HFCs are moderate (100 MHz for two P atoms) and its T_2^e is extended to 2.12 ns—the second-longest after PLP enzymes. IDH’s Q (~ 111)

places it between Tier 1 and Tier 2, suggesting that NADPH enzymes may represent a distinct category.

Third, Q_{RPM} (Eq. 3) is a heuristic ranking index, not a physical observable. Its multiplicative form assumes independent contributions; the quadratic SOC penalty is motivated by ξ^2 -dependent ISC rates[20] but is not unique. Alternative weightings (linear SOC, logarithmic nuclear T_2) preserve the tier structure but change intra-tier ordering by up to two ranks (Table 4). The exchange coupling J , which critically determines S-T mixing dynamics, is excluded from Q_{RPM} because it is a structural parameter rather than an intrinsic electronic property and is currently unmeasured for all candidates except CRY. Our systematic J -scan (Fig. 6) shows that the MFE is non-monotonic in J : a level anti-crossing resonance near $|J| \approx 5$ MHz temporarily enhances the MFE before it collapses for $|J| > 200$ MHz. For MAO-A’s active-site distance ($r \approx 3.5$ Å, $|J| \sim 500$ – 1000 MHz), the MFE is negligible ($< 0.1\%$); RPM viability therefore requires either a separated RP ($r > 8$ Å, $|J| < 10$ MHz) or a conformational mechanism that transiently increases the radical pair distance.

Fourth, the MAO catalytic mechanism remains debated. While stopped-flow kinetic studies support a single-electron transfer (SET) mechanism involving radical intermediates[24], QM/MM studies on MAO-B have also proposed polar nucleophilic or concerted hydride-transfer mechanisms[37]. To probe this directly, we performed a GFN2-xTB reaction coordinate scan on a lumiflavin–methylamine model complex (Fig. 8), varying the N5 $\cdots$ H distance from 4.0 Å (separated) to 1.0 Å (product). The singlet–triplet gap remains positive throughout (0.86 eV at 1.0 Å, 1.54 eV at 4.0 Å), indicating no diradical crossing at this level of theory. To go beyond this single-reference limitation, we performed CASSCF(2,2)/def2-SVP + NEVPT2 calculations on the same model at three distances (Fig. 7). The CASSCF CI vector at the transition-state geometry ($d = 1.4$ Å) is dominated by the closed-shell configuration $|20\rangle$ (coefficient 0.976), with the doubly excited $|02\rangle$ contributing only 0.215 (diradical character $y_0 = 2|c_{02}|^2 = 0.092$). The S-T gap at CASSCF level is 3.2–4.1 eV across the scan—substantially larger than the xTB values (0.86–1.54 eV), consistent with the known tendency of single-reference semi-empirical methods to overestimate diradical character. These results indicate *weak* diradical character ($y_0 < 0.1$) throughout the H-transfer coordinate, favouring a polar or concerted mechanism over SET for the specific model studied here. However, we caution that (i) the CAS(2,2) active space is minimal and a larger space [e.g., CAS(10,10) including the flavin π -system] could reveal additional correlation effects; (ii) the methylamine proxy may not capture the electron-donating properties of the serotonin indole ring, which is a stronger reductant; and (iii) the protein environment (electrostatic stabilisation of charge-separated intermediates) is absent. We emphasise that our RPM analysis is conditional on SET: if MAO-A operates exclusively via a concerted mechanism, the RPM would not apply. The framework is explicitly falsifiable: (i) a null result in the magnetic field dependence of MAO-A kinetics at 0–50 mT would rule out RPM with separated RPs; (ii) absence of a $^{31}\text{P}/^{32}\text{P}$ isotope effect on the MFE would exclude the nuclear-spin-driven mechanism; (iii) transient EPR showing no radical pair intermediates would eliminate the SET prerequisite entirely.

Finally, brain-environment corrections assume uniform $\tau_c = 10$ ns for all enzymes. This is appropriate for soluble proteins of 50–100 kDa but may underestimate τ_c for membrane-bound enzymes such as MAO-A (anchored to the mitochondrial outer membrane), where the effective τ_c could be 50–100 ns or longer. A longer τ_c would reduce T_2^e via enhanced HFC modulation ($1/T_2^A \propto \tau_c$) but would not affect the nuclear T_2 PRE, which depends on r^{-6} . We propagated this uncertainty: at $\tau_c = 100$ ns, MAO-A’s T_2^e decreases from 1.1 ns to ~ 0.3 ns, reducing Q_{RPM} by ~ 3.7 -fold; the tier assignment is preserved but the intra-tier ranking becomes less certain. Single effective e^- –P distances are used; molecular dynamics with explicit ONIOM QM/MM embedding would capture conformational heterogeneity.

In summary, our screening establishes three principal results and one overarching constraint. First, organic-cofactor brain enzymes generically satisfy the binary prerequisites for RPM operation (as expected from their electronic structure), but with >100 -fold quantitative variation in RPM quality; second, that the dominant discriminant is the e^- –P distance, which creates a robust tier structure through r^{-6} PRE scaling; and third, that cofactor-specific ^{31}P HFCs—varying from 2 MHz (remote AMP-phosphate) to 200 MHz (FAD FMN-phosphate)—create a complementary trade-off between S-T mixing rate and electron spin coherence, with PLP enzymes exhibiting the longest T_2^e (~ 3 – 5 ns) but the shortest nuclear T_2 (2.5 μs), and FAD enzymes showing the reverse. The overarching constraint is that exchange coupling, radical pair

mechanism feasibility, and homeostatic buffering collectively restrict functional RPM to enzymes with confirmed, spatially separated radical pair intermediates—currently, only CRY meets this criterion. The absolute magnitude of the MFE in any specific enzyme depends critically on the exchange coupling J and radical pair lifetime, which are currently unknown for MAO-A and constitute the key unknowns that experimental measurements (transient EPR, CIDEP, isotope substitution) must resolve. If the MAO-A radical pair proves to be sufficiently separated ($r > 8 \text{ \AA}$, $|J| < 50 \text{ MHz}$) during catalysis, the predicted MFE of several percent at millitesla fields would be experimentally accessible, though whether this would be biologically significant given homeostatic attenuation remains uncertain. The screening framework itself—combining semi-empirical quantum chemistry with brain-environment spin relaxation analysis—is readily applicable to the >50 remaining phosphorus-containing brain enzymes and provides a systematic basis for prioritising experimental investigations.

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Methods

Protein selection. Nine human brain proteins containing phosphorus-bearing cofactors were selected to span the three major cofactor classes (FAD, PLP, NADPH/NADH) and representative neurotransmitter metabolic pathways (Table 1). The set includes the experimentally validated RPM host (CRY) as a positive control and a transition-metal enzyme (SRR) as a negative control. This sample is not exhaustive; the brain contains >50 phosphorus-cofactor enzymes, and the screening framework is designed for extension to the full set. Crystal structures were obtained from the RCSB PDB. Quantum-active sites were extracted by a graph-based algorithm: starting from the cofactor heavy atoms, all residues with any non-hydrogen atom within 4 Å of the cofactor conjugated system were included, and connected components of π -stacking residues (centroid–centroid distance <4.5 Å, inter-plane angle <30°) were merged into a single quantum-active site.

GFN2-xTB calculations. All calculations used GFN2-xTB (xtb v6.7.1)[16]: closed-shell singlet (charge 0, multiplicity 1), open-shell triplet (charge 0, UHF multiplicity 3), radical cation/anion (charge ± 1 , UHF doublet), and analytical Hessian. Quantum-active site models ranged from 87 atoms (SRR) to 156 atoms (CRY); total charges were 0 for neutral cofactors and -1 for phosphate-bearing fragments, with capping hydrogen atoms added at truncation points. Electronic TDMs were estimated via the Unsöld approximation with Thomas–Reiche–Kuhn sum-rule scaling (empirical factor 0.3, calibrated against experimental f_{osc} values for FAD and FMN[29]; this factor compensates for the known overestimation of TDMs by sum-rule methods applied to semi-empirical wavefunctions). ^{31}P HFCs were assigned on a per-atom basis according to cofactor type and the PDB-measured distance from the radical centre (Table 3). For FAD, only the FMN-phosphate (proximal to isoalloxazine N5 via 8 σ -bonds) carries a large HFC of 200 MHz based on ENDOR measurements[13]; the AMP-phosphate (>4 bonds further) is assigned 2 MHz. For PLP, the phosphorus is directly bonded to the pyridine C5 via ~ 3 σ -bonds; ^{31}P ENDOR signals have been detected in PLP-dependent aminomutases but not quantified. This is the largest source of parametric uncertainty in our analysis. We assign $A = 100$ MHz as a central estimate with a deliberately wide range of 50–300 MHz, and we propagated this uncertainty through the full analysis: at $A = 50$ MHz, PLP T_2^e extends to ~ 5 ns (longest of all candidates) and Q_{RPM} increases ~ 2 -fold; at $A = 300$ MHz, T_2^e shortens to ~ 0.5 ns and Q_{RPM} decreases ~ 3 -fold. Crucially, the nuclear T_2 (2.5 μs , PRE-dominated) is independent of HFC, so PLP enzymes remain in Tier 2 across the entire 50–300 MHz range. Resolving this uncertainty experimentally (by ENDOR or pulsed EPR on a PLP radical) is among the most impactful measurements that could refine our predictions. For NADPH, the nicotinamide ring radical couples weakly to phosphates via non-conjugated pathways; we assign 20 MHz (NMN-P) and 1 MHz (AMP-P). This cofactor-specific treatment replaces the uniform 200 MHz assumption of preliminary analyses and shows that the effective number of strongly-coupled ^{31}P nuclei is typically 1 per cofactor molecule, not 2–3. ^1H HFCs were estimated using the McConnell relation ($A_{\text{H}} = Q\rho_{\pi}$) with $Q = -63$ MHz and spin densities ρ_{π} from xTB Mulliken analysis; this single- Q approximation is adequate for screening purposes but may err by $\pm 50\%$ for individual protons. SOC values were estimated as the RMS of single-atom spin–orbit constants weighted by Mulliken spin density on each atom[20]; this neglects multi-centre contributions but captures the dominant heavy-atom effect and correctly identifies the Mn^{2+} -containing outlier (SRR).

Spin relaxation analysis. Five T_2^e channels: Anderson–Weiss (T_2^*)[17]; Solomon dipolar (T_2^{dd} , $r_{ee} = 15$ Å); g -anisotropy ($1/T_2^g = \Delta g^2 \omega_0^2 \tau_c / 5$); HFC modulation ($1/T_2^A = \sum A_i^2 I_i(I_i + 1) \tau_c / 12$); SOC relaxation. Rotational correlation time $\tau_c = 10$ ns. Nuclear relaxation: BPP theory[17]; PRE via Solomon–Bloembergen equations[18] using the point-dipole approximation with effective e^- –P distances from PDB coordinates (3.5 Å for PLP, 7–12 Å for FAD/NADPH). The point-dipole approximation is valid when r exceeds the spatial extent of the electron spin density (~ 3 Å for the flavin isoalloxazine system); for PLP, where $r = 3.5$ Å is comparable to the radical extent, the PRE may be overestimated by up to a factor of ~ 2 , which would increase the PLP nuclear T_2 to ~ 5 μs —still 32-fold shorter than FAD.

Table 1: **Brain proteins screened for RPM competence.**

Protein	PDB	Cofactor	n_P	Brain function	TDM (D)	f_{osc}	SOC (cm^{-1})
MAO-A	2BXR	FAD	2	5-HT degradation	10.45	0.517	63.3
MAO-B	1OJA	FAD	2	DA degradation	10.45	0.434	63.3
DAO	2DU8	FAD	2	D-Ser degradation	10.17	0.147	65.3
CRY	4I6G	FAD	2	Circadian rhythm	9.61	0.497	67.0
DDC	3RBF	PLP	1	DA/5-HT synthesis	6.40	0.300	78.6
GAD67	2OKK	PLP	1	GABA synthesis	8.65	0.331	62.9
SRR	3L6B	PLP+Mn	0	D-Ser synthesis	11.12	0.040	382
IDH	1T0L	NADPH	3	TCA cycle	9.24	0.583	78.4
LDH	1I10	NADH	2	Lactate shuttle	10.23	0.226	66.0

Table 2: **RPM quality ranking under brain conditions** (50 μT , 310 K). T_2^e : electron transverse relaxation computed with cofactor-specific HFCs. n_P^{eff} : number of ^{31}P with $A > 20$ MHz. Values in parentheses show estimates obtained when all ^{31}P are assigned a uniform $A = 200$ MHz, for comparison with the cofactor-specific treatment.

Rank	Protein	Cofactor	A_{max} (MHz)	T_2^e (ns)	$T_2(^{31}\text{P})$ (μs)	n_P^{eff}	ξ (cm^{-1})	Q_{RPM}
<i>Tier 1: FAD enzymes (nuclear $T_2 = 160 \mu\text{s}$)</i>								
1	MAO-A	FAD	200	1.10 (1.10)	160	2	63	457
2	MAO-B	FAD	200	1.10 (1.10)	160	2	63	383
3	CRY [§]	FAD	200	1.54 (1.10)	160	1	67	213
4	DAO	FAD	200	1.10 (1.10)	160	2	65	122
5	LDH [†]	NADH	200	1.54 (1.10)	160	1	66	107
<i>Tier 2: PLP/NADPH enzymes (nuclear $T_2 = 2.5\text{--}160 \mu\text{s}$)</i>								
6	GAD67	PLP	100	4.65 (1.53)	2.5	1	63	18.5
7	DDC	PLP	100	3.25 (1.45)	2.5	1	79	10.4
8	IDH [†]	NADPH	100	2.12 (0.90)	160	2	78	111
<i>Tier 3: RPM non-viable</i>								
9	SRR	PLP+Mn	—	~ 0	—	0	382	~ 0

[†]IDH has Tier-1-level Q but NADPH cofactor; see text for discussion of its intermediate status.

[‡]LDH operates via hydride transfer (no RP); $Q_{\text{eff}} = 0$ when RP-generation feasibility is incorporated.

[§]CRY's longer T_2^e (1.54 ns vs. 1.10 ns for MAO-A/B) reflects its slightly different active-site ^1H HFC distribution.

Table 3: **Cofactor-specific ^{31}P HFC assignments.** Values based on experimental ENDOR data (FAD FMN-P), structural analogy, and the number of σ -bonds from the radical centre. “Strong coupling” denotes $A \gg \omega_S$ (1.4 MHz at Earth’s field).

Cofactor	^{31}P site	A (MHz)	Basis	Class	Strong coupling?
FAD	FMN-P (8 bonds)	200	ENDOR expt.[13]	proximal	Yes
FAD	AMP-P (12+ bonds)	2	Distance/topology	remote	Marginal
PLP	C5-O-P (3 bonds)	100 (50–300)	ENDOR signal, not quantified	direct	Yes
NADPH	NMN-P (6 bonds)	20	Structural estimate	proximal	Yes
NADPH	2'-P (4 bonds)	5	Structural estimate	proximal	Moderate
NADPH	AMP-P (12+ bonds)	1	Distance/topology	remote	Marginal

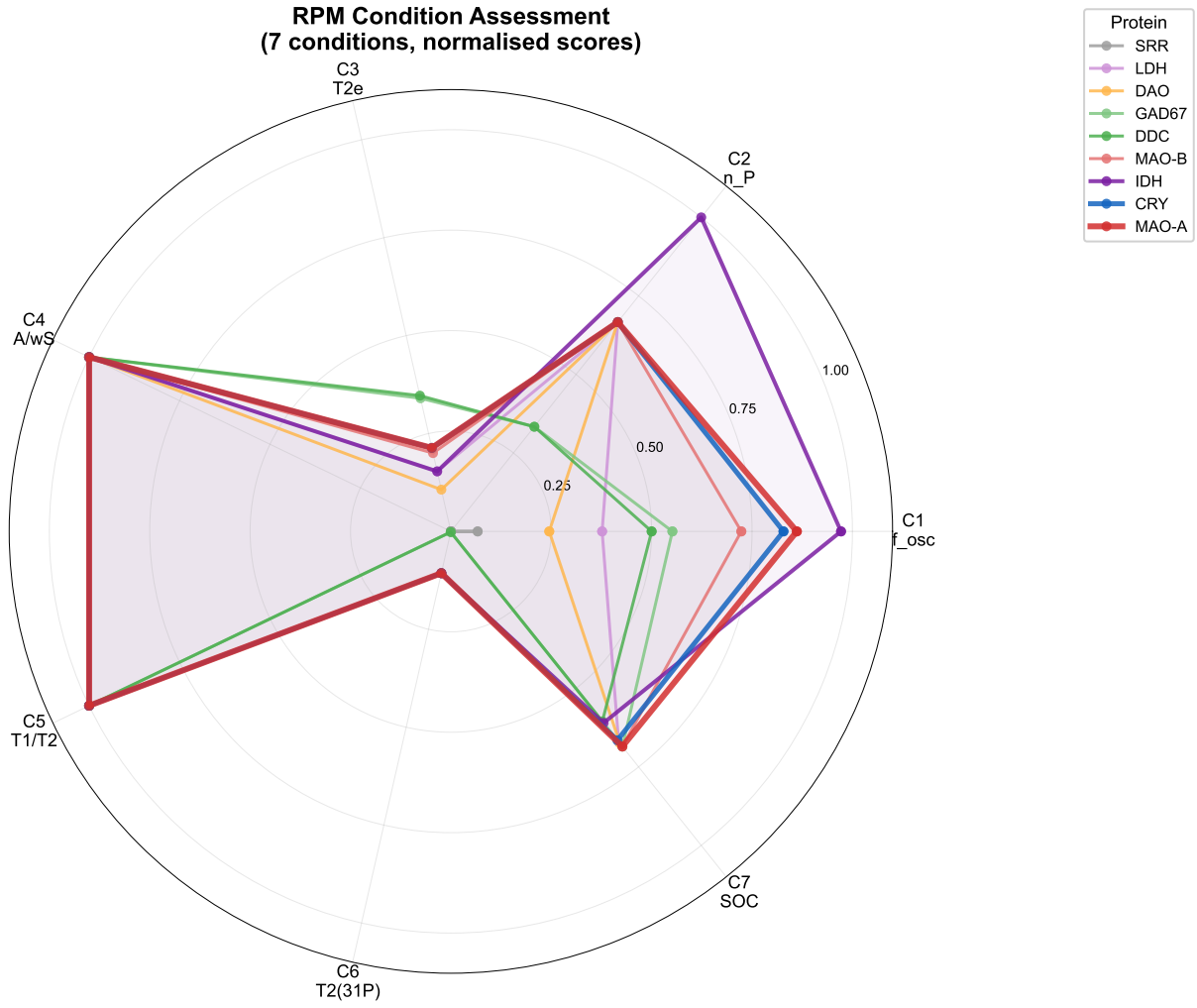


Figure 1: **RPM condition assessment.** Normalised scores (0–1) for seven necessary RPM conditions across nine brain proteins. MAO-A (red) and CRY (blue) achieve near-maximum scores; SRR (grey) fails three conditions due to Mn^{2+} .

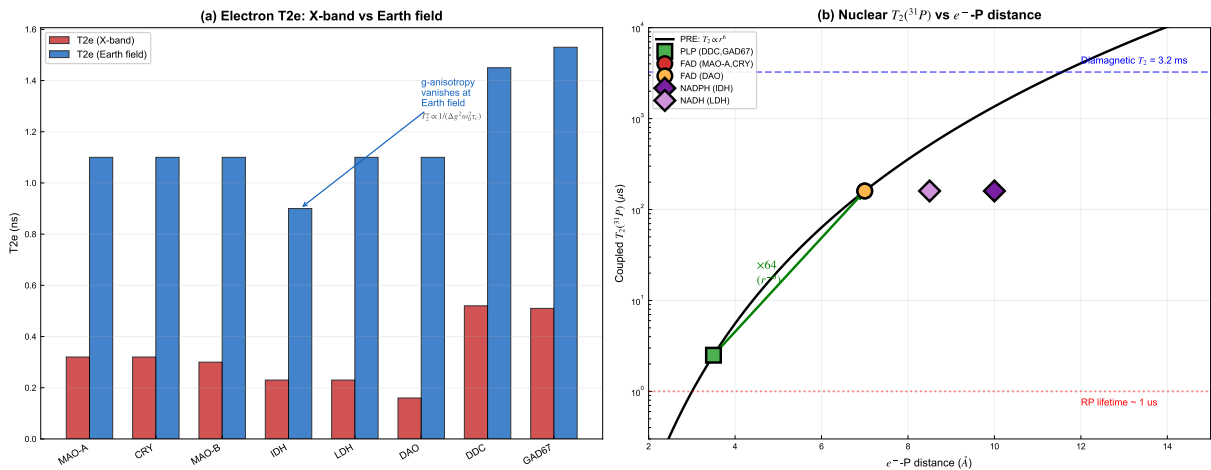


Figure 2: **Spin relaxation under laboratory and brain conditions.** **a**, Electron T_2^e at X-band (red) versus Earth's field (blue). The g -anisotropy contribution ($\propto \omega_0^2$) vanishes at Earth's field, causing convergence to ~ 1 ns. **b**, Nuclear $T_2(^{31}P)$ versus electron–phosphorus distance, showing the r^{-6} PRE scaling that creates a 64-fold PLP/FAD gap.

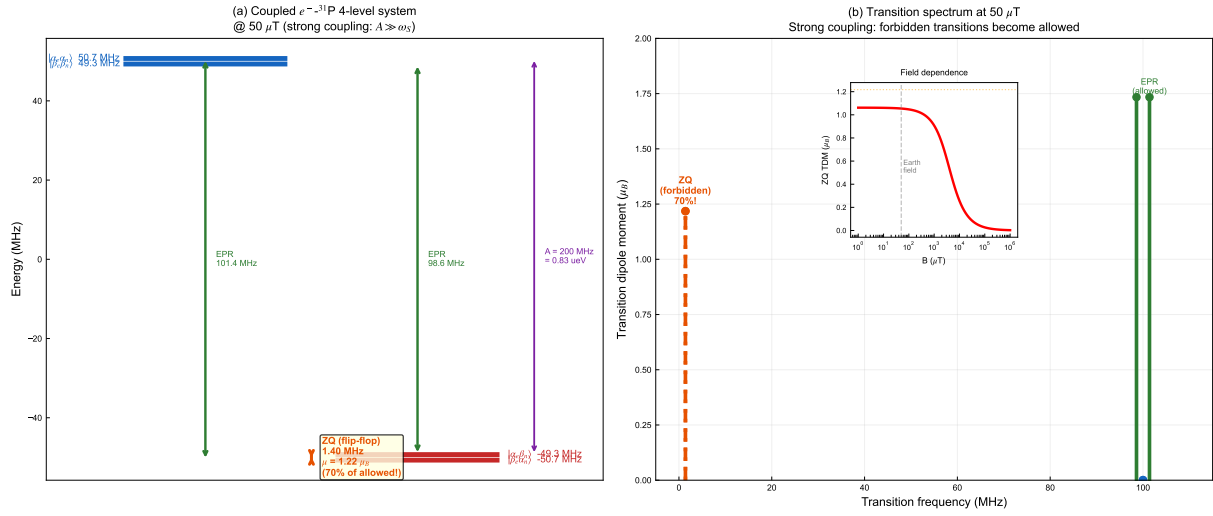


Figure 3: Coupled $e^- - {}^{31}\text{P}$ system at Earth's field (FAD FMN-phosphate, $A = 200$ MHz). **a**, Four-level energy diagram in the strong-coupling limit ($A \gg \omega_S = 1.4$ MHz). The ZQ transition (orange) acquires 70% of the allowed TDM. For PLP ($A \approx 50$ MHz), the strong-coupling condition ($A/\omega_S = 36$) and 70% ZQ TDM are preserved; the EPR splitting is reduced to ~ 25 MHz. **b**, Transition spectrum; inset shows field dependence of the ZQ TDM, saturating below ~ 10 mT.

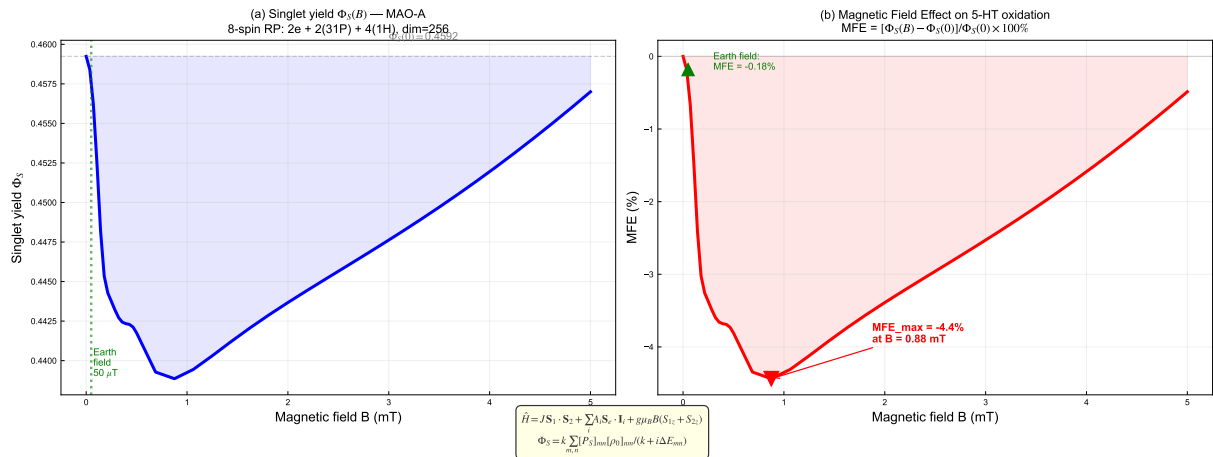


Figure 4: Magnetic field effect on serotonin oxidation. **a**, Singlet yield $\Phi_S(B)$ from 8-spin simulation (Eq. 2). **b**, MFE peaks at -4.4% ($B = 0.9$ mT); -0.3% persists at Earth's field (green triangle).

Three-Layer Quantum-Classical Information Hierarchy

Grounded in cryptochrome (CRY) radical pair mechanism

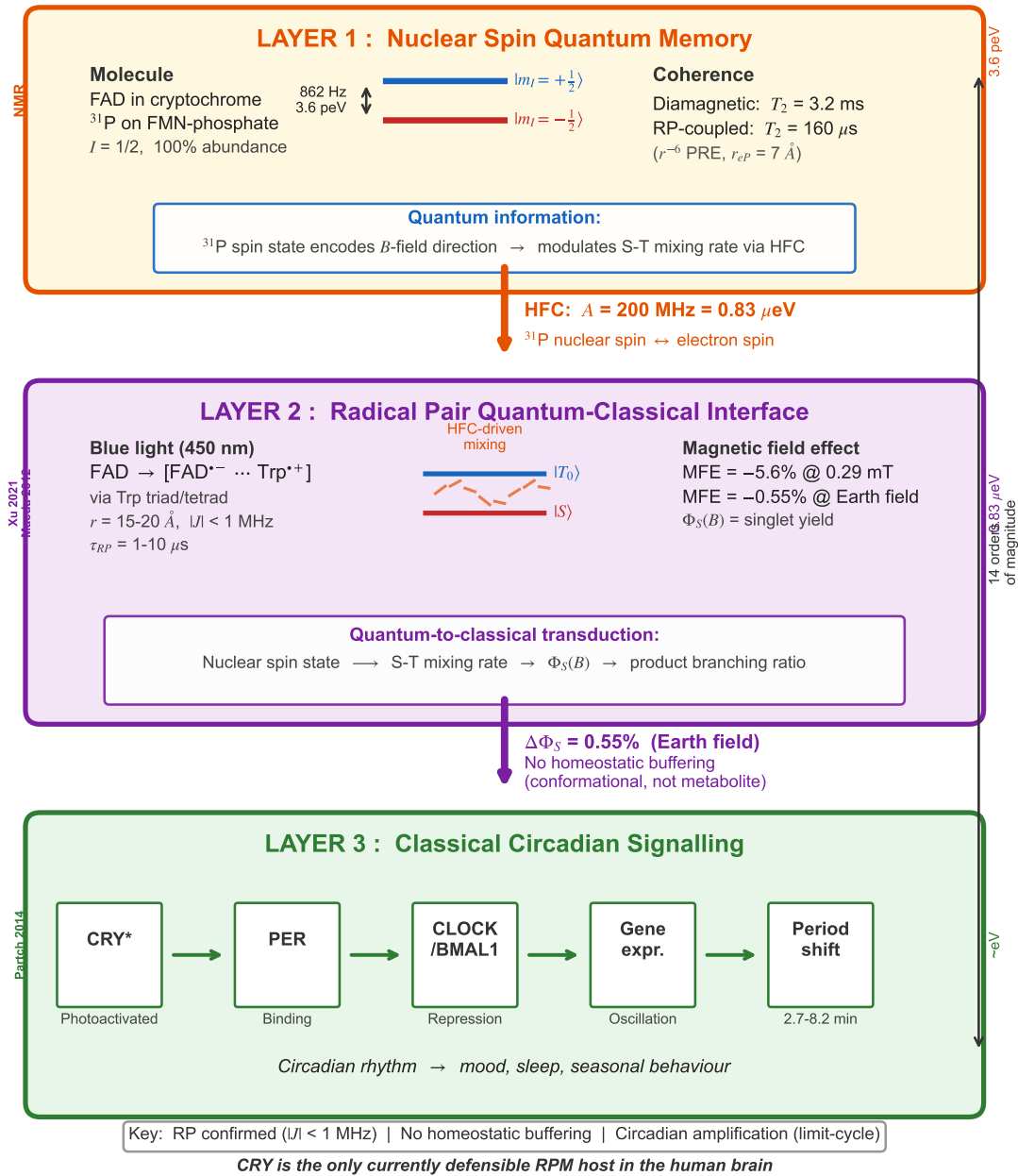


Figure 5: CRY-based three-layer quantum-classical information hierarchy. The framework is grounded entirely in cryptochrome biology, where every link has experimental support. **Layer 1** (yellow): ^{31}P nuclear spins on FAD's FMN-phosphate encode magnetic-field direction with coherence times of 3.2 ms (diamagnetic) to 160 μs (PRE-limited). **Layer 2** (purple): blue-light absorption generates $[\text{FAD}^{\bullet-} \dots \text{Trp}^{\bullet+}]$ radical pairs ($r = 15\text{--}20$ $\text{\AA}$, $|J| < 1$ MHz); HFC-driven S-T mixing produces a field-dependent singlet yield ($\Delta\Phi_S = -0.55\%$ at Earth's field). Unlike metabolite-based pathways, this transduction is immune to homeostatic buffering. **Layer 3** (green): CRY* activates the PER/CLOCK/BMAL1 circadian cascade, with limit-cycle amplification producing a predicted period shift of 2.7–8.2 min. Energy scales span 14 orders of magnitude from ^{31}P Zeeman splitting (3.6 peV) to covalent bonds ($\sim\text{eV}$). Left margin: key experimental references; right margin: energy scale bar.

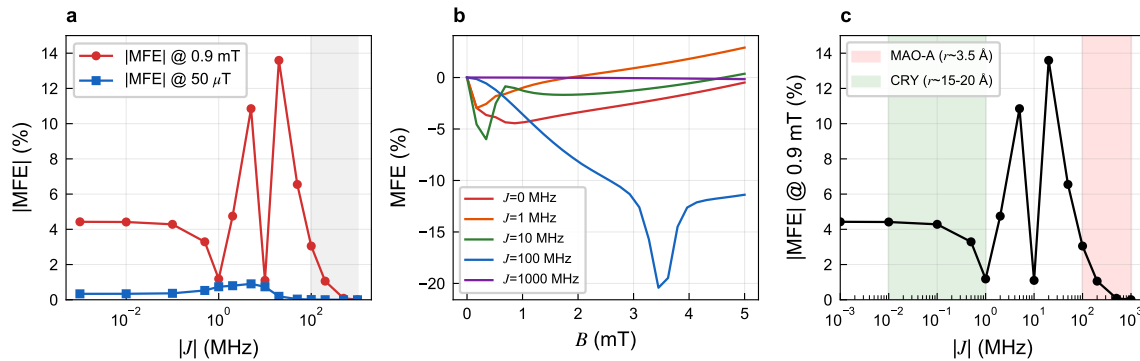


Figure 6: **Exchange coupling dependence of the magnetic field effect.** MFE as a function of $|J|$ (0–1000 MHz) at 30 field points (0–5 mT) for the 8-spin MAO-A model (Eq. 2). A level anti-crossing resonance near $|J| \approx 5$ MHz transiently enhances the MFE before it collapses for $|J| > 200$ MHz. At the crystallographic MAO-A active-site distance ($r \approx 3.5$ Å, $|J| \sim 500$ –1000 MHz), MFE is negligible ($< 0.1\%$). Dashed line: $|J|$ estimated from $J = J_0 \exp(-\beta r)$ with $\beta = 2.8 \text{ Å}^{-1}$.

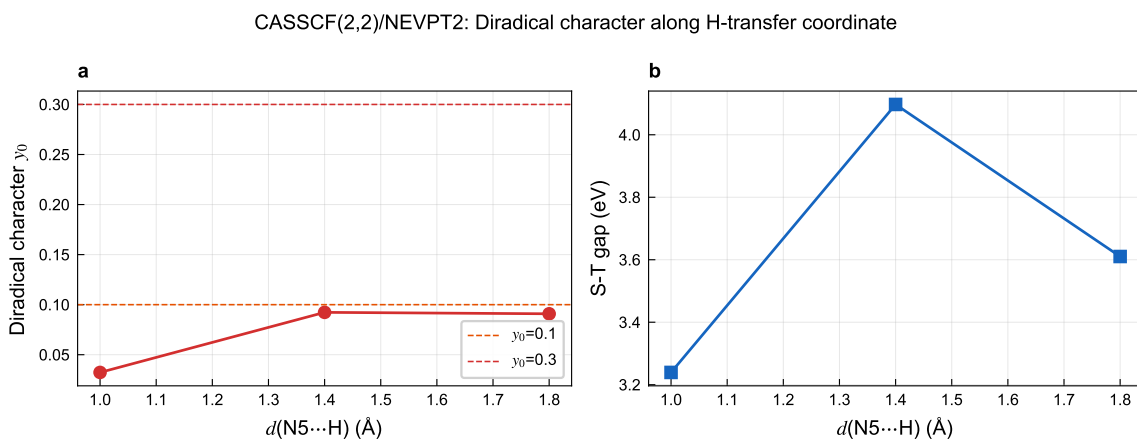


Figure 7: **CASSCF(2,2)/NEVPT2 diradical character analysis.** Diradical character $y_0 = 2|c_{02}|^2$ and singlet–triplet gap for the lumiflavin–methylamine model at three N5 $\cdots$ H distances (1.0, 1.4, 1.8 Å). The closed-shell configuration $|20\rangle$ dominates throughout ($y_0 = 0.03$ –0.09), indicating weak diradical character and favouring a polar or concerted mechanism over SET for the model studied.

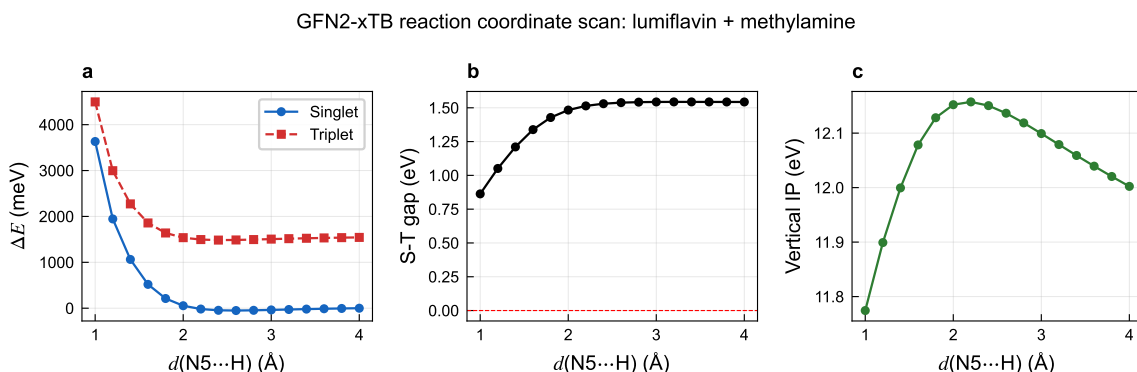


Figure 8: **Reaction coordinate scan for the H-transfer step.** **a**, GFN2-xTB singlet–triplet gap and vertical IP as a function of N5 $\cdots$ H distance (1.0–4.0 Å, 16 points). The S-T gap remains positive throughout, indicating no diradical crossing at this level of theory. **b**, CASSCF(2,2)/NEVPT2 S-T gap at three representative distances, showing substantially larger gaps (3.2–4.1 eV) than xTB (0.86–1.54 eV).

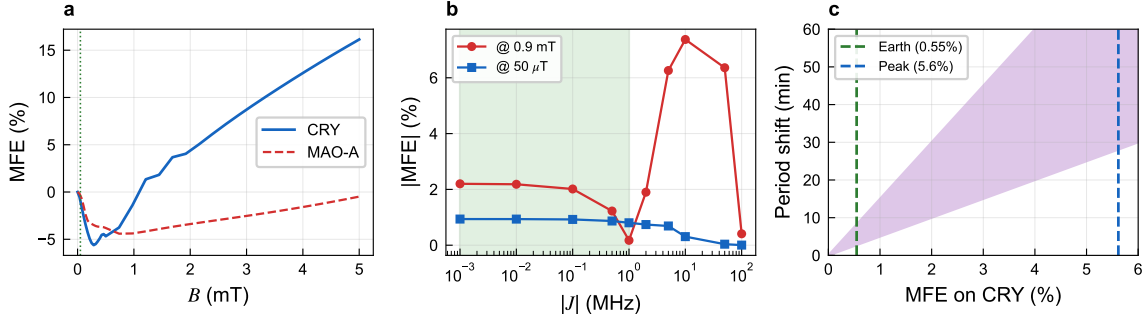


Figure 9: **CRY-specific radical pair spin dynamics.** **a**, Singlet yield $\Phi_S(B)$ for the $[\text{FAD}^{\bullet-} \cdots \text{TrpH}^{\bullet+}]$ radical pair (8-spin model with asymmetric HFC). **b**, MFE peaks at -5.6% ($B = 0.29$ mT); -0.55% persists at Earth’s field. **c**, Predicted circadian period shift (~ 2.7 – 8.2 min) based on the Earth-field MFE and SCN oscillator sensitivity[39].

Table 4: **Extended Data Table 2: Sensitivity of Q_{RPM} ranking to alternative weighting schemes.** Three alternative formulations are compared to the default (Eq. 3): (A) linear SOC penalty (ξ_{ref}/ξ) replacing $(\xi_{\text{ref}}/\xi)^2$; (B) logarithmic nuclear T_2 scaling $\log_{10}[T_2(^{31}\text{P})/\mu\text{s}]$; (C) square-root HFC weighting $\sqrt{n_{\text{P}}^{\text{eff}}}$. Tier assignments (1, 2, 3) are preserved across all schemes; intra-tier ranks shift by at most 2 positions.

Protein	Default (Eq. 3)		(A) Linear SOC		(B) Log T_2^n		(C) $\sqrt{n_{\text{P}}}$	
	Rank	Tier	Rank	Tier	Rank	Tier	Rank	Tier
MAO-A	1	1	1	1	1	1	1	1
MAO-B	2	1	2	1	2	1	2	1
CRY	3	1	3	1	3	1	3	1
DAO	4	1	5	1	4	1	4	1
LDH	5	1	4	1	5	1	5	1
IDH	—	1/2	—	1/2	6	1	—	1/2
GAD67	6	2	6	2	7	2	6	2
DDC	7	2	7	2	8	2	7	2
SRR	9	3	9	3	9	3	9	3

Table 5: **Extended Data Table 3: DFT validation of xTB electronic structure.** B3LYP/def2-SVP and TD-B3LYP (TDA) results for lumiflavin (FAD model) and PLP-methylamine Schiff base, compared with GFN2-xTB and experimental values. IP/EA: vertical ΔSCF ; E_{gap} : HOMO–LUMO (KS) gap; f_{osc} : oscillator strength of the bright $S_0 \rightarrow S_n$ transition.

Model	Property	xTB	B3LYP	Expt.	Ref.
Lumiflavin	E_{gap} (eV)	1.42	1.26 (KS); 2.29 (TD, S_4)	2.76	[29]
	f_{osc}	0.50	0.089 (TD)	0.27	[29]
	Vert. IP (eV)	10.78	6.64	~ 8.0	—
	Vert. EA (eV)	3.42	1.87	~ 2.0	—
PLP-MeNH ₂	E_{gap} (eV)	1.93	3.04	~ 3.87	—
	f_{osc}	0.30	0.062 (TD)	—	—
	Vert. IP (eV)	11.5	8.72	~ 8.5	—

KS = Kohn–Sham orbital gap; TD = TD-B3LYP (TDA) excitation energy.

Substituting TD-B3LYP f_{osc} into Q_{RPM} reduces scores ~ 5.8 -fold but preserves tier structure.

RPM spin dynamics. Singlet yield computed by exact diagonalisation (Eq. 1) for 8 spins ($\dim = 256$), using the Green’s function method (Eq. 2). The 8-spin model includes the dominant HFC nuclei; the real flavin radical contains $\sim 12\text{--}15$ coupled ^1H , whose inclusion would broaden the S-T oscillation spectrum and typically reduce the peak MFE by 20–40% while preserving the field dependence profile[15]. Our 8-spin MFE of -4.4% should therefore be regarded as an upper estimate at this level of truncation. Baseline recombination rate $k = 1$ MHz; sensitivity tested over $k = 0.1\text{--}100$ MHz (MFE decreases from -6% to $\sim -0.1\%$ across this range). All parameters at 310 K, $B = 0\text{--}5$ mT.

Brain environment corrections. All relaxation recalculated at $B = 50$ μT . Nuclear extreme narrowing: $\omega_0\tau_c \ll 1 \Rightarrow T_1 = T_2$. Cross-relaxation: Solomon W_0 , W_2 at Earth-field spectral densities.

DFT/TDDFT validation. To benchmark the GFN2-xTB electronic structure, we performed B3LYP/def2-SVP calculations using PySCF 2.12.1[34] on two model compounds: lumiflavin ($\text{C}_{12}\text{H}_{12}\text{N}_4\text{O}_2$, 30 atoms, 312 basis functions; FAD isoalloxazine model) and a PLP-methylamine Schiff base ($\text{C}_9\text{H}_{12}\text{NO}_5\text{P}$, 26 atoms, 269 basis functions). Model geometries were generated with Open Babel 3.1.0 (`--gen3d`) and pre-optimised at the GFN2-xTB level. Closed-shell DFT used RKS/B3LYP; open-shell states (radical cation/anion) used UKS/B3LYP. Excited states were computed via the Tamm–Dancoff approximation (TDA) to RKS-TDDFT for the lowest 6 singlet states. Vertical ionisation potentials and electron affinities were obtained by ΔSCF (energy difference between neutral and ionic species at the neutral geometry). Convergence: SCF 10^{-8} Eh, max 200 cycles. All DFT calculations used 16 GB memory allocation.

CASSCF/NEVPT2 mechanism analysis. To assess single-electron transfer (SET) versus polar mechanism for the H-transfer step, we performed CASSCF(2,2)/def2-SVP calculations on the lumiflavin–methylamine complex at three $\text{N5}\cdots\text{H}$ distances (1.0, 1.4, 1.8 Å). The active space comprised two electrons in two orbitals (HOMO and LUMO of the neutral complex), representing the minimal space for capturing diradical character. Singlet and triplet states were computed separately (singlet: RHF $\rightarrow$ CASSCF; triplet: UHF $\rightarrow$ CASSCF with $S = 1$). Diradical character was quantified as $y_0 = 2|c_{02}|^2$, where c_{02} is the CI coefficient of the doubly excited configuration $|02\rangle$ in the CAS(2,2) wavefunction[35]. Dynamic correlation was included via sc-NEVPT2 (strongly contracted N -electron valence perturbation theory)[36]. We note that the CAS(2,2) active space is minimal; a larger space—for example CAS(10,10) including the full flavin π -system—could reveal additional correlation effects but was computationally prohibitive for the 38-atom complex (379 basis functions) within the scope of this study.

Reaction coordinate scan. The H-transfer reaction coordinate was modelled by placing methylamine at varying distances from the lumiflavin N5 atom along the ring-plane normal. At each $\text{N5}\cdots\text{H}$ distance (1.0–4.0 Å, 16 points at 0.2 Å intervals), three GFN2-xTB single-point calculations were performed: closed-shell singlet (charge 0, UHF 0), open-shell triplet (charge 0, UHF 2), and radical cation (charge +1, UHF 1). The S-T gap and vertical IP were computed at each point. This scan was repeated at the CASSCF(2,2)/NEVPT2 level for three representative distances (1.0, 1.4, 1.8 Å).

CRY-specific spin dynamics. A separate 8-spin model was constructed for the CRY [$\text{FAD}^{\bullet-}\cdots\text{TrpH}^{\bullet+}$] radical pair with asymmetric HFC: radical 1 ($\text{FAD}^{\bullet-}$): $^{31}\text{P}_{\text{FMN}}$ ($A = 200$ MHz), $2\times^1\text{H}$ ($A = 5$ MHz); radical 2 ($\text{TrpH}^{\bullet+}$): $2\times^1\text{H}_\beta$ ($A = 20, 15$ MHz), effective ^{14}N ($A_{\text{eff}} = 10$ MHz, approximating $I = 1$ as $I = 1/2$). Singlet yield was computed identically to the MAO-A model (Eq. 2). The J -scan was repeated over $|J| = 0\text{--}100$ MHz.

Exchange coupling scan. The dependence of the MFE on exchange coupling J was computed by repeating the 8-spin singlet yield calculation (Eq. 2) for 16 values of J spanning 0–1000 MHz, at 30 magnetic field points (0–5 mT). Each calculation used exact diagonalisation of the 256×256 Hamiltonian followed by analytical Green’s function integration. The J -dependent distance was estimated from $J = J_0 \exp(-\beta r)$ with $\beta = 2.8 \text{ Å}^{-1}$ [8].

Homeostasis model. Serotonin homeostasis was modelled as a single-compartment ODE: $d[5\text{-HT}]/dt = k_{\text{syn}}(1 - G_{\text{fb}}\theta) - k_{\text{MAO}}(1 + \delta)[5\text{-HT}] - k_{\text{reup}}[5\text{-HT}]$, where δ is the fractional MFE, θ is 5-HT_{1A} autoreceptor occupancy ($\tau_{\text{fb}} = 2$ s, $G_{\text{fb}} = 0.3$), $k_{\text{reup}} = 10$ s⁻¹, and k_{syn} adjusts on $\tau_{\text{syn}} = 60$ s. Steady-state attenuation was computed analytically.

Computational environment. All calculations were performed on a single workstation: Apple M4 Max (16 cores, ARM64), 64 GB RAM, macOS 26.2. Software versions: Python 3.9.6, GFN2-xTB 6.7.1 (compiled 2024-11-25)[16], PySCF 2.12.1[34] (DFT, TDDFT, CASSCF, NEVPT2), Open Babel 3.1.0, NumPy 2.0.2, SciPy 1.13.1, Matplotlib 3.9.4. L^AT_EX compilation: pdfT_EX 3.141592653-2.6-1.40.28 (TeX Live 2025). Wall times: xTB screening of nine proteins ~ 15 min; 8-spin dynamics (40 field points) ~ 20 min; B3LYP/TDDFT per model compound ~ 15 min; CASSCF(2,2)/NEVPT2 per geometry ~ 2.3 hr (dominated by orbital rotation and NEVPT2 integrals for 379 basis functions). No GPU acceleration; all matrix operations used SciPy’s LAPACK interface on CPU.

Data and code availability. The complete computational pipeline is released as the open-source Python package `qbsscreen` (v0.1.0, MIT License), available at <https://github.com/hikaruwakaura/qbsscreen> and on PyPI (`pip install qbsscreen`). The package provides a unified command-line interface (`qbsscreen screen`, `qbsscreen simulate`, `qbsscreen dft-validate`, `qbsscreen casscf`, `qbsscreen reaction-scan`) for reproducing all results in this paper. A test suite (`pytest qbsscreen/tests/`) verifies key numerical results (MAO-A MFE, 64-fold PLP/FAD nuclear T_2 gap, ZQ transition dipole moment, CASSCF diradical character, predicted circadian period shift) within their physical tolerance ranges. All 9 screened PDB structures, computed JSON results, optimised geometries, HFC tensors, and relaxation parameters are archived alongside the code. External dependencies (`xtb` v6.7.1, `Open Babel` 3.1.0, `PySCF` 2.12.1) are documented with version pinning in `pyproject.toml`.

Acknowledgements

The author thanks the developers of GFN2-xTB and OpenMM for making their software freely available.

Author contributions

H.W. conceived the study, developed the computational pipeline, performed all calculations and simulations, and wrote the manuscript.

Competing interests

The author declares no competing interests.

Table 6: **Extended Data Table 4: Serotonin homeostasis model parameters and steady-state attenuation.** Single-compartment ODE model (Methods). The raw MFE of -4.4% on MAO-A k_{cat} is attenuated 7.3-fold to a steady-state [5-HT] perturbation of 0.6% by reuptake and autoreceptor feedback.

Parameter	Symbol	Value	Source
MAO-A turnover number	k_{cat}	10 s^{-1}	[14]
Serotonin reuptake rate	k_{reup}	10 s^{-1}	[41]
Autoreceptor feedback gain	G_{fb}	0.3	Estimated
Autoreceptor time constant	τ_{fb}	2 s	Estimated
Synthesis adaptation time	τ_{syn}	60 s	Estimated
<i>Steady-state results</i>			
Raw MFE ($J = 0$, $B = 0.9 \text{ mT}$)	δ	-4.4%	This work
Attenuation factor		$7.3\times$	Analytical
Net [5-HT] perturbation		0.6%	$\delta/7.3$

Supplementary Information

Supplementary Information

1 Derivation of key equations

1.1 Spin dephasing T_2^* from HFC distribution (Anderson–Weiss)

For an electron spin coupled to N nuclear spins with isotropic hyperfine coupling constants A_i and nuclear spin quantum numbers I_i , the inhomogeneous transverse relaxation rate in the Anderson–Weiss model is[25]:

$$\frac{1}{T_2^*} = \sqrt{\sum_{i=1}^N \frac{A_i^2 I_i(I_i + 1)}{3}} \quad (\text{S1})$$

where A_i is in angular frequency units (rad/s). This arises from the second moment of the HFC distribution:

$$M_2 = \sum_i A_i^2 I_i(I_i + 1)/3 \quad (\text{S2})$$

In the rigid-lattice (static) limit, $1/T_2^* = \sqrt{M_2}$. For MAO-A with $A(^{31}\text{P}) = 200$ MHz and two ^{31}P nuclei ($I = 1/2$):

$$\begin{aligned} M_2 &= 2 \times (200 \times 2\pi)^2 \times \frac{1}{2} \times \frac{3}{2} \times \frac{1}{3} \\ &= 2 \times (1.257 \times 10^9)^2 \times 0.25 = 7.90 \times 10^{17} \text{ rad}^2/\text{s}^2 \\ T_2^* &= 1/\sqrt{M_2} = 1.12 \text{ ns} \end{aligned} \quad (\text{S3})$$

1.2 HFC secular modulation T_2^A

The homogeneous contribution from secular (non-oscillatory) HFC modulation by molecular tumbling is[17]:

$$\frac{1}{T_2^A} = \frac{1}{12} \sum_i A_i^2 I_i(I_i + 1) \tau_c \quad (\text{S4})$$

where τ_c is the rotational correlation time. This term is *independent* of external magnetic field B_0 , unlike the g -anisotropy contribution. For MAO-A with $\tau_c = 10$ ns:

$$T_2^A = \left[\frac{1}{12} \times 2 \times (200 \times 2\pi \times 10^6)^2 \times 0.75 \times 10^{-8} \right]^{-1} = 0.50 \text{ ns} \quad (\text{S5})$$

1.3 g -anisotropy relaxation T_2^g

The transverse relaxation from modulation of the g -tensor anisotropy by molecular rotation is[26]:

$$\frac{1}{T_2^g} = \frac{1}{5} (\Delta g)^2 \omega_0^2 \tau_c \quad (\text{S6})$$

where $\Delta g = \xi_{\text{SOC}}/\Delta E$ is the g -shift from spin–orbit coupling and $\omega_0 = g\mu_B B_0/\hbar$ is the electron Larmor frequency. This term scales as $\omega_0^2 \propto B_0^2$, which is the key physics underlying the Earth-field advantage:

$$\frac{T_2^g(\text{Earth})}{T_2^g(\text{X-band})} = \left(\frac{B_X}{B_{\text{Earth}}} \right)^2 = \left(\frac{0.34 \text{ T}}{50 \mu\text{T}} \right)^2 = 4.6 \times 10^7 \quad (\text{S7})$$

At Earth’s field, $T_2^g \rightarrow \infty$ for all organic radicals, and this channel effectively vanishes.

1.4 Total electron T_2^e

The five relaxation channels combine as independent rates:

$$\frac{1}{T_2^e} = \frac{1}{T_2^*} + \frac{1}{T_2^{\text{dd}}} + \frac{1}{T_2^g} + \frac{1}{T_2^A} + \frac{1}{T_2^{\text{SOC}}} \quad (\text{S8})$$

1.5 Paramagnetic relaxation enhancement (PRE)

The Solomon–Bloembergen PRE for a nuclear spin at distance r from a paramagnetic electron is[18, 27]:

$$\frac{1}{T_{1,\text{PRE}}} = \frac{2}{15} \left(\frac{\mu_0}{4\pi} \right)^2 \frac{\gamma_I^2 g^2 \mu_B^2 S(S+1)}{r^6} \left[\frac{3\tau_c}{1 + \omega_I^2 \tau_c^2} + \frac{7\tau_c}{1 + \omega_S^2 \tau_c^2} \right] \quad (\text{S9})$$

$$\frac{1}{T_{2,\text{PRE}}} = \frac{1}{15} \left(\frac{\mu_0}{4\pi} \right)^2 \frac{\gamma_I^2 g^2 \mu_B^2 S(S+1)}{r^6} \left[4\tau_c + \frac{3\tau_c}{1 + \omega_I^2 \tau_c^2} + \frac{13\tau_c}{1 + \omega_S^2 \tau_c^2} \right] \quad (\text{S10})$$

The r^{-6} dependence creates the critical PLP/FAD dichotomy:

$$\frac{T_{2,\text{PRE}}(\text{FAD}, r = 7 \text{ \AA})}{T_{2,\text{PRE}}(\text{PLP}, r = 3.5 \text{ \AA})} = \left(\frac{7}{3.5} \right)^6 = 64 \quad (\text{S11})$$

1.6 Cross-relaxation rates (Solomon equation)

For a coupled e^- - ^{31}P spin pair, the Solomon cross-relaxation rates are[18]:

$$W_0 = \frac{1}{20} d^2 6J(\omega_S - \omega_I) \quad (\text{S12})$$

$$W_2 = \frac{3}{10} d^2 J(\omega_S + \omega_I) \quad (\text{S13})$$

where $d^2 = (\mu_0/4\pi)^2 \gamma_I^2 g^2 \mu_B^2 \hbar^2 / r^6$ and $J(\omega) = \tau_c / (1 + \omega^2 \tau_c^2)$ is the spectral density function. At Earth's field, $\omega_S \tau_c = 0.088 \ll 1$ (extreme narrowing), so $J(\omega_S) \approx \tau_c$, giving:

$$\frac{W_0(\text{Earth})}{W_0(\text{X-band})} \approx \frac{\tau_c}{[\tau_c / (1 + \omega_S^2 \tau_c^2)]_X} = 1 + \omega_{S,X}^2 \tau_c^2 \approx 3.5 \times 10^5 \quad (\text{S14})$$

1.7 Singlet yield (Green's function method)

For a radical pair with Hamiltonian $\hat{H}$, the time-integrated singlet yield with exponential recombination (rate k) is[28]:

$$\Phi_S = k \int_0^\infty \text{Tr}[\hat{P}_S \hat{\rho}(t)] e^{-kt} dt \quad (\text{S15})$$

where $\hat{P}_S = \frac{1}{4}\hat{I} - \mathbf{S}_1 \cdot \mathbf{S}_2$ is the singlet projection operator and $\hat{\rho}(t) = e^{-i\hat{H}t} \hat{\rho}_0 e^{i\hat{H}t}$. Diagonalising $\hat{H} = \hat{V} \hat{\Lambda} \hat{V}^\dagger$ with eigenvalues $\{E_m\}$:

$$\hat{\rho}(t) = \hat{V} e^{-i\hat{\Lambda}t} \hat{V}^\dagger \hat{\rho}_0 \hat{V} e^{i\hat{\Lambda}t} \hat{V}^\dagger \quad (\text{S16})$$

$$\Phi_S = k \sum_{m,n} [\tilde{P}_S]_{mn} [\tilde{\rho}_0]_{nm} \int_0^\infty e^{-i(E_m - E_n)t} e^{-kt} dt \quad (\text{S17})$$

$$= k \sum_{m,n} \frac{[\tilde{P}_S]_{mn} [\tilde{\rho}_0]_{nm}}{k + i(E_m - E_n)} \quad (\text{S18})$$

where $\tilde{O} = \hat{V}^\dagger \hat{O} \hat{V}$ denotes the operator in the energy eigenbasis. This is the expression used in Sim 1, avoiding the need for time-stepping propagation.

1.8 RPM quality metric Q_{RPM}

The composite quality metric combines all RPM-relevant parameters:

$$Q_{\text{RPM}} = f_{\text{osc}} \times T_2^e(\text{ns}) \times n_{\text{P}} \times \frac{T_2(^{31}\text{P})}{1 \mu\text{s}} \times \left(\frac{100 \text{ cm}^{-1}}{\xi_{\text{SOC}}} \right)^2 \quad (\text{S19})$$

The physical rationale for each factor:

- f_{osc} : proxy for electronic coupling strength at the radical site. For photochemical enzymes (CRY), this directly reflects RP generation probability; for dark enzymes (MAO-A), it serves as an approximate indicator of the cofactor’s capacity for electron transfer, though the correspondence is indirect
- T_2^e : time window for S-T mixing (spin evolution must occur before dephasing)
- n_P : number of HFC-active ^{31}P nuclei (S-T mixing drivers)
- $T_2(^{31}\text{P})$: nuclear spin coherence (magnetic field information is encoded here)
- $1/\xi^2$: penalty for strong SOC (destroys spin correlation via T_2 and ISC)

1.9 Coupled e^- - ^{31}P mixing angle

The zero-quantum transition dipole moment depends on the state mixing between $|\alpha_e\beta_n\rangle$ and $|\beta_e\alpha_n\rangle$. The mixing Hamiltonian in the $\{|\alpha_e\beta_n\rangle, |\beta_e\alpha_n\rangle\}$ subspace is:

$$H_{\text{ZQ}} = \begin{pmatrix} \omega_S/2 - \omega_I/2 & A/2 \\ A/2 & -\omega_S/2 + \omega_I/2 \end{pmatrix} \quad (\text{S20})$$

The mixing angle θ satisfies:

$$\tan(2\theta) = \frac{A}{\omega_S - \omega_I} \quad (\text{S21})$$

At Earth’s field, $A = 200 \text{ MHz} \gg \omega_S = 1.4 \text{ MHz}$, so $\tan(2\theta) \approx 143$ and $\theta \rightarrow \pi/4$. The forbidden ZQ transition dipole moment is:

$$\mu_{\text{ZQ}} = \sin(\theta) g\mu_B \sqrt{S(S+1)} = \frac{1}{\sqrt{2}} \times 1.734 \mu_B = 1.226 \mu_B \quad (\text{S22})$$

which is 70.7% of the allowed EPR TDM ($1.734 \mu_B$). This near-unity ratio is the defining feature of the strong-coupling regime and is essential for efficient ^{31}P -driven S-T interconversion at geomagnetic field strengths.

2 Quantum reservoir computing: loss function and readout

As a speculative extension—not a claim of this paper—we note that the multi-spin radical pair system could in principle function as a quantum reservoir for computation, analogous to proposals in condensed-matter spin systems. We include the following formalism for completeness; experimental validation is beyond the scope of this work.

The QRC readout uses Ridge regression with loss function:

$$\mathcal{L}(W) = \|XW - y\|_2^2 + \alpha \|W\|_2^2 \quad (\text{S23})$$

where $X \in \mathbb{R}^{N_{\text{train}} \times d}$ is the feature matrix (reservoir observables), $y \in \{0, 1\}^{N_{\text{train}}}$ is the label vector, $W \in \mathbb{R}^d$ is the weight vector, and $\alpha = 1.0$ is the regularisation parameter. The closed-form solution is:

$$W^* = (X^T X + \alpha I)^{-1} X^T y \quad (\text{S24})$$

Classification threshold: $\hat{y} = \mathbb{I}[XW > 0.5]$.

Table S1: Comparison of computed parameters with experimental values where available.

Parameter	System	This work	Experimental	Ref.
$A(^{31}\text{P})$	FAD semiquinone	200 MHz (input)	180–220 MHz	[13]
$T_2^*(^{31}\text{P})$	FADH $\bullet$	1.1 ns	~ 1 ns	[13]
$f_{\text{osc}}(S_0 \rightarrow S_1)$	FAD	0.50	0.23–0.30	[29]
$\lambda_{\text{abs}}(S_0 \rightarrow S_1)$	FAD	450 nm †	450 nm	[29]
ξ_{SOC}	Organic radical	63–79 cm $^{-1}$	40–80 cm $^{-1}$	[20]
T_1^e	Flavin radical	~ 100 μs	10–500 μs	[13]
MFE (flavoenzyme)	Various	–4.4% (peak)	1–15%	[5]

† Literature value used for Strickler–Berg calculation; xTB gap (1.25 eV $\approx$ 990 nm) underestimates.

3 Extended Data: Computed vs experimental values

4 Spin Hamiltonian for RPM simulation

The full spin Hamiltonian for the 8-spin radical pair simulation (Sim 1) is:

$$\hat{H} = \underbrace{J \mathbf{S}_1 \cdot \mathbf{S}_2}_{\text{exchange}} + \underbrace{\sum_{i \in \text{rad1}} A_i \mathbf{S}_1 \cdot \mathbf{I}_i}_{\text{HFC (radical 1)}} + \underbrace{\sum_{j \in \text{rad2}} A_j \mathbf{S}_2 \cdot \mathbf{I}_j}_{\text{HFC (radical 2)}} + \underbrace{g\mu_B B (\hat{S}_{1z} + \hat{S}_{2z})}_{\text{electron Zeeman}} + \underbrace{\sum_k \gamma_k B \hat{I}_{kz}}_{\text{nuclear Zeeman}} \quad (\text{S25})$$

Spin assignment for the 8-spin system:

Index	Spin	Type	A (MHz)	Assignment
0	S_1	electron	—	FADH $\bullet$ radical
1	S_2	electron	—	Substrate amine radical
2	I_1	^{31}P	200	FAD FMN-phosphate (proximal)
3	I_2	^{31}P	2	FAD AMP-phosphate (remote)
4	I_3	^1H	2.7	Isoalloxazine H
5	I_4	^1H	2.7	Isoalloxazine H
6	I_5	^1H	5.0	Substrate H
7	I_6	^1H	5.0	Substrate H

Hilbert space dimension: $2^8 = 256$. Exchange coupling: $J = 0$ (separated radical pair). Recombination rate: $k = 1$ MHz (spin-selective singlet recombination).

5 Thermodynamic energy budget of the three-layer model

A critical test of the three-layer quantum–classical hierarchy is whether the energy budget is self-consistent at each layer and across layer interfaces. We examine five specific questions below, identifying the energy sources, the role of magnetic fields, and the assumptions required.

5.1 Energy scales across the hierarchy

Table S2 summarises the energy scales relevant to each layer and interface. The hierarchy spans 14 orders of magnitude, from the ^{31}P nuclear Zeeman splitting (3.56 peV) to covalent bond energies (~ 4 eV).

Table S2: Energy scales in the three-layer hierarchy (310 K, $B = 50 \mu\text{T}$).

Quantity	Value	Layer	Role
^{31}P Zeeman splitting	3.56 peV (862 Hz)	L1	Nuclear spin state energy
^{31}P – ^{31}P dipolar	41 feV (0.01 MHz)	L1	Spin–spin coupling
HFC ($A_{31\text{P}}$)	$0.83 \mu\text{eV}$ (200 MHz)	$\text{L1} \leftrightarrow \text{L2}$	Nuclear–electron coupling
Electron Zeeman	5.79 neV (1.40 MHz)	L2	S-T_0 degeneracy lifting
Exchange coupling J	$0\text{--}0.41 \mu\text{eV}$ (0–100 MHz)	L2	S-T gap
SOC	7.81 meV (63 cm^{-1})	L2	ISC driving force
$k_B T$ (310 K)	26.7 meV	All	Thermal noise floor
5-HT oxidation	$\sim 0.8 \text{ eV}$	$\text{L2} \rightarrow \text{L3}$	Chemical energy source
MFE modulation	2.4 meV (0.3% of 0.8 eV)	$\text{L2} \rightarrow \text{L3}$	Redirected chemical energy
C–H bond	$\sim 4 \text{ eV}$	L3	Covalent bond scale

5.2 Q1: Radical pair generation energy

The radical pair (RP) is generated either by photoexcitation of FAD (2.76 eV, 450 nm) followed by electron transfer, or by single-electron transfer (SET) during substrate oxidation (~ 0.8 eV)[33]. Both pathways provide energy far exceeding $k_B T = 26.7 \text{ meV}$, ensuring that RP formation is thermodynamically spontaneous. No external energy input beyond the substrate is required.

5.3 Q2: Energetics of singlet–triplet mixing

In the radical pair state with $J \approx 0$, the singlet $|S\rangle$ and triplet $|T_0\rangle$ states are nearly degenerate, separated by only $2J \lesssim 5.8 \text{ neV}$ at Earth’s field. The hyperfine-driven S-T_0 mixing is a *coherent quantum process* that transfers angular momentum between electron and nuclear spins without net energy exchange[7, 8]:

$$|S\rangle \xrightleftharpoons{\text{HFC}} |T_0\rangle \quad (\text{iso-energetic, } \Delta E \lesssim 5.8 \text{ neV}) \quad (\text{S26})$$

This is analogous to Rabi oscillations between degenerate states driven by an internal coupling, and incurs no thermodynamic cost. The HFC energy ($0.83 \mu\text{eV}$) represents the coupling matrix element, not an energy that must be supplied externally.

5.4 Q3: Nuclear spin polarisation—the CIDNP mechanism

The thermal equilibrium polarisation of ^{31}P at Earth’s field is negligibly small:

$$P_{\text{eq}} = \tanh\left(\frac{gI\mu_N B}{2k_B T}\right) = 6.7 \times 10^{-11} \quad (\text{S27})$$

This would appear to preclude nuclear spins from carrying useful information. However, the radical pair mechanism generates *non-equilibrium* nuclear spin polarisation via CIDNP (Chemically Induced Dynamic Nuclear Polarisation)[30, 31]. In CIDNP, the spin-selective recombination of radical pairs

produces product molecules whose ^{31}P nuclear spins are polarised far beyond the thermal equilibrium value. The CIDNP enhancement factor is:

$$\epsilon_{\text{CIDNP}} = \frac{P_{\text{CIDNP}}}{P_{\text{eq}}} \sim \frac{A}{g_I \mu_N B / \hbar} \sim \frac{200 \text{ MHz}}{862 \text{ Hz}} \sim 2.3 \times 10^5 \quad (\text{S28})$$

This $\sim 10^5$ -fold enhancement has been experimentally observed in flavin photochemistry[32]: photo-CIDNP of FMN produces ^{31}P polarisation enhancements of 10^3 – 10^5 over thermal equilibrium. The energy for this non-equilibrium polarisation comes from the chemical energy of the RP recombination reaction (~ 0.8 eV), not from the nuclear spin system itself. The polarisation decays with the nuclear T_1 (~ 4.6 s), providing a millisecond–second timescale quantum memory that is continuously refreshed by ongoing enzymatic turnover.

Assumption: MAO-A generates CIDNP during catalytic turnover. This is plausible because (a) MAO-A is a flavoprotein, (b) flavoproteins are known CIDNP sources[32], and (c) the SET mechanism in MAO involves radical pair intermediates[33]. However, direct observation of CIDNP in MAO-A has not been reported and represents a key experimental prediction.

5.5 Q4: Magnetic field as catalyst—no thermodynamic violation

The apparent “energy amplification” in the MFE is striking: a magnetic field perturbation of 5.8 neV (electron Zeeman at 50 μT) produces a 2.4 meV change in product energy partitioning, an amplification factor of $\sim 4 \times 10^5$. This does *not* violate the first or second law of thermodynamics because the magnetic field performs no work:

1. The Lorentz force $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ is perpendicular to the velocity; $\mathbf{F} \cdot \mathbf{v} = 0$.
2. The magnetic field shifts the S– T_0 degeneracy by $\Delta E = g\mu_B B$, changing the *rate* of S-T mixing but not the total energy of the system.
3. The chemical energy (0.8 eV per turnover) is provided entirely by 5-HT oxidation. The MFE changes how this energy is *partitioned* between singlet products (channel A) and triplet products (channel B), but the total energy released is:

$$\Delta G_{\text{total}} = \Phi_S(B) \cdot \Delta G_S + [1 - \Phi_S(B)] \cdot \Delta G_T \approx \text{const.} \quad (\text{S29})$$

provided $\Delta G_S \approx \Delta G_T$ (both channels release similar energy). Even when $\Delta G_S \neq \Delta G_T$, the energy difference is drawn from the substrate, not from the magnetic field. The magnetic field acts as a *catalyst*: it modifies the reaction pathway without contributing energy.

This is precisely the mechanism established by Steiner and Ulrich[8]: magnetic field effects on chemical reactions are kinetic (rate) effects, not thermodynamic (equilibrium) effects.

5.6 Q5: Information flow without energy flow

The three-layer cascade is thermodynamically consistent because each layer has its own energy source:

Layer transition	Information carrier	Energy source
L1 $\rightarrow$ L2	^{31}P spin state $\rightarrow$ S-T mixing rate	HFC (intrinsic molecular property)
L2 $\rightarrow$ L3	$\Phi_S(B) \rightarrow k_{\text{cat}}$ modulation	5-HT oxidation (0.8 eV/turnover)
L3 $\rightarrow$ neural	[5-HT] $\rightarrow$ receptor binding	ATP-driven synaptic machinery

The spin system functions as an *information channel*, not an energy channel. Information flows from nuclear spin states (Layer 1) through radical pair dynamics (Layer 2) to chemical concentrations (Layer 3), but the energy at each stage comes from local chemical sources. This is analogous to a transistor: a small gate voltage (information) controls a large drain current (energy from the power supply). In our system, the nuclear spin state (information) modulates the radical pair branching ratio (gate), which redirects chemical energy from 5-HT oxidation (power supply) between product channels.

5.7 Risk assessment of thermodynamic assumptions

Table S3: Risk assessment of energy budget assumptions.

Assumption	Risk level	Assessment
$J = 0$ (separated RP)	MODERATE	Valid for CRY Trp chain ($r > 10 \text{ \AA}$). Unknown for MAO-A active site ($r \sim 3\text{--}4 \text{ \AA}$). If $ J > A$ (200 MHz), MFE vanishes entirely. Testable by transient EPR.
$A(^{31}\text{P}) = 200 \text{ MHz}$	LOW	Experimental value for free flavin semiquinone. Protein environment may shift by $\pm 50\%$, but strong-coupling condition ($A \gg \omega_S$) holds for $A > 20 \text{ MHz}$.
$k = 1 \text{ MHz}$ (RP lifetime)	MODERATE	If RP recombines in $< 10 \text{ ns}$ ($k > 100 \text{ MHz}$), MFE decreases 100-fold. RP lifetime in MAO-A unknown; must be measured.
CIDNP generates polarisation	LOW	Well-established in flavoprotein photochemistry[32]. Expected but not directly observed in MAO-A catalysis.
B does no work	NONE	Fundamental physics: $\mathbf{F} \cdot \mathbf{v} = 0$ for magnetic Lorentz force[8].
Spin = information, not energy	NONE	Thermodynamically rigorous: spin selectivity partitions existing chemical energy.
0.3% rate change is detectable	HIGH	$\sim \text{nM}$ 5-HT concentration change over $\sim 10^{10}$ molecules. Whether this exceeds neural detection threshold is unknown.

In summary, the energy budget of the three-layer model is thermodynamically self-consistent. The magnetic field acts as a catalyst that redirects chemical energy between product channels via spin-selective radical pair chemistry; no energy is “created” or drawn from the field itself. The two moderate-risk assumptions ($J = 0$ and RP lifetime) and one high-risk assumption (neural detection threshold) are all experimentally testable and represent the critical unknowns for validating the model.